



EduFocus

A senior project submitted in partial fulfillment of the requirements for the degree
of Bachelor of Computers and Artificial Intelligence.

“Computer Science” Program,

Presented by

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Benha, June 2026

ACKNOWLEDGMENT

The success of any project depends largely on the encouragement and guidelines of many others. We take this opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project.

The special thanks go to our helpful supervisor, **Dr.Mai Kamal**.

And support she gave truly help in the progression of our graduation project. The co-operation is much indeed appreciated.

Our greatly appreciate the contribution of our faculty, Faculty of Computer and Artificial Intelligence (BFC AI).

Our grateful thanks also go to **Eng. Mawada Ashraf**.

A significant contribution and hard work from both of you during the last ten months. The project would be nothing without enthusiasm, help and support from both of you. Besides, this project makes us realize the value of working together as a team and as new experience in the working environment, which challenges us every minute.

DECLARATION

We hereby certify that this material, which we now submit for assessment on the program of study leading to the award of Bachelor of Computers and Informatics in **Computer Science** is entirely our own work, that we have exercised reasonable care to ensure that the work is original, and does not to the best of our knowledge breach any law of copyright, and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of our work .

Signed: _____

Date: _____

Abstract

Digital education remains largely inaccessible for a significant cohort of students who face the intersection of cognitive and physical barriers, specifically those managing both attention deficit hyperactivity disorder (ADHD) and severe motor impairments. **EduFocus Adaptive** was developed to bridge this divide, driven by the principle of "radical inclusion" the belief that technology must adapt to the unique needs of the user, not the other way around.

The platform introduces a "dual engine" approach to personalized learning. To address physical barriers, the system utilizes a computer vision-based control system that enables full application navigation and interaction entirely through eye-tracking, eliminating the need for mouse, keyboard, or touch inputs. To address cognitive engagement, an AI Adaptive Lesson Engine generates and dynamically adjusts interactive content in real-time, tailoring the difficulty and style of lessons based on the student's focus and performance.

Built on a robust architecture using Flutter and on-device TensorFlow Lite models, the application prioritizes privacy by processing all biometric eye-tracking data locally. By creating a closed-loop environment where a child's gaze and cognition are the only prerequisites for interaction, EduFocus Adaptive aims to foster scholastic independence, build confidence, and deliver a truly frustration-free educational experience.

INTRODUCTION

EduFocus Adaptive represents a next-generation approach to digital education, specifically engineered to transcend the "one-size-fits-all" model that currently dominates the market. The project is driven by a philosophy of "radical inclusion," aiming to serve a specific but underserved cohort of students: children navigating the intersection of cognitive challenges, such as ADHD, and severe physical motor impairments.

For these students, standard educational tools present insurmountable barriers. The static nature of traditional software fails to engage neurodivergent minds, while physical interfaces like mice and keyboards effectively exclude those with conditions like Cerebral Palsy. To bridge this gap, EduFocus Adaptive utilizes a "dual-engine" system:

1. **AI Adaptive Lesson Engine:** A dynamic system that generates and adjusts content in real-time based on the user's focus and performance, ensuring the material remains engaging.
2. **Computer Vision Control:** A gaze-based interaction model that enables full navigation entirely through eye movements, eliminating the need for physical touch.

By creating this self-contained, autonomous learning loop, the platform ensures that a child's focus and cognition are the only prerequisites for success, fostering scholastic independence and confidence.

PROBLEM DEFINITION

The core problem driving this project is that digital learning tools remain profoundly inaccessible for a significant cohort of students. This inaccessibility is not singular; it exists as a "double burden" where traditional educational software fails to accommodate two distinct but often co-occurring needs:

- **The Cognitive Barrier (Focus & Engagement):** Standard educational software relies on static content and rigid progression models that fail to retain the focus of neurodivergent students, particularly those with ADHD. The lack of dynamic stimulation in current tools leads to disengagement, frustration, and a perception of failure, which ultimately widens the learning gap.
- **The Physical Barrier (Accessibility):** The vast majority of learning applications are architected around specific physical inputs—mouse, keyboard, or touchscreen. For children with conditions such as Cerebral Palsy, Muscular Dystrophy, or spinal cord injuries, these interfaces represent an insurmountable barrier that effectively excludes them from the modern digital classroom.

The Critical Gap While assistive technologies exist for physical disabilities, and gamified apps exist for ADHD, a critical gap remains at the intersection of these two challenges. Few, if any, solutions provide a holistic environment for a child who faces both cognitive and physical hurdles simultaneously.

IMPORTANCE OF THIS PROBLEM

The urgency of this project stems from a critical failure in modern educational technology: the "one-size-fits-all" model leaves behind students who face complex, intersecting challenges. The importance of solving this can be categorized into three distinct areas:

1. The "Double Barrier" to Education

Current digital learning environments inadvertently punish students who deviate from the norm, creating a cycle of failure for those with dual diagnoses.

- **The Cognitive Cost:** For neurodivergent students (e.g., ADHD), the standard static content and rigid progression of educational software fail to maintain engagement. This lack of dynamic stimulation creates a "perception of failure" and actively widens the learning gap.
- **The Physical Cost:** For students with motor impairments (e.g., Cerebral Palsy), the reliance on mouse, keyboard, or touchscreen inputs acts as an "insurmountable barrier". These students are not just struggling; they are effectively "locked out" of modern digital learning solely due to physical limitations.

2. The Gap of Intersectionality

While the market offers solutions for ADHD (gamification) and solutions for physical accessibility (assistive hardware), a massive void exists where these two needs intersect.

- There are few, if any, holistic solutions for a child who faces *both* cognitive and physical hurdles.
- Without a tool like EduFocus Adaptive, these students are forced to use fragmented tools that do not accommodate their full reality, leading to disjointed and ineffective learning experiences.

3. The Imperative of "Radical Inclusion"

The project is motivated by the ethical principle that technology must adapt to the user, rather than forcing the user to adapt to the technology.

- **Restoring Independence:** Currently, these students often require constant external help to navigate interfaces. Solving this problem creates a "closed-loop" system where the student can manage their learning independently—from calibration to lesson completion.
- **Building Confidence:** By replacing frustration with an environment where "focus and cognition are the only prerequisites," the solution aims to rebuild the student's confidence and scholastic self-worth.



PROBLEM SOLUTION

Overview:

To achieve the vision of "radical inclusion," the platform moves beyond standard accessibility patches. Instead, it provides a holistic, self-contained environment where a student's focus and cognition are the only prerequisites for interaction.

The solution is composed of two primary engines working in tandem:

1. Solving the Physical Barrier: Computer Vision Control System

To address the needs of students with motor impairments (e.g., Cerebral Palsy), the system removes the reliance on traditional hardware inputs like mice, keyboards, or touchscreens.

- **Zero-Touch Interface:** The application enables 100% of navigation and interaction to be performed entirely through a webcam-based eye-tracking system.
- **Dwell-to-Click Interaction:** Selection is handled by a "dwell" mechanism, where holding a gaze on a target triggers a click, visualized by a closing ring indicator.
- **On-Device Processing:** To ensure privacy and speed, all computer vision processing occurs locally on the device, with no video feeds sent to external servers.

2. Solving the Cognitive Barrier: AI Adaptive Lesson Engine

To address the engagement challenges faced by neurodivergent students (e.g., ADHD), the system replaces static content with dynamic, responsive learning paths.

- **Real-Time Adaptation:** The AI engine generates and adjusts lessons on the fly based on the user's live performance and focus levels.

- **Gamified Micro-Lessons:** Content is broken down into short, 3-5 minute "missions" to maintain high engagement and provide rapid feedback.
- **Focus-Aware "Nudges":** Rather than punishing distraction, the system uses gaze tracking to detect "wandering." If a student looks away for too long, the system provides a gentle, non-punitive audio-visual "nudge" to re-engage them.

The Outcome: An Autonomous Learning Loop

By integrating these two engines, EduFocus Adaptive creates a "closed-loop" system. This allows the student to independently manage their entire session—from the initial eye-tracking calibration to the completion of the lesson—without the need for external physical assistance, fostering a sense of autonomy and confidence.

WHAT'S NEW?

The primary innovation is the shift from a "one-size-fits-all" model to a "**dual-engine**" system that solves two distinct accessibility problems simultaneously.

1. The "Radical Inclusion" Philosophy

Unlike standard tools that force users to adapt to the software, this system is built on the principle that the technology must adapt to the user.

- **Novelty: It addresses a critical gap at the intersection of cognitive (ADHD) and physical (motor impairment) challenges, offering a holistic solution where few currently exist**

2. A "Zero-Touch" Interaction Model

Most educational apps require physical interaction (mouse/touch). EduFocus Adaptive introduces a completely hands-free experience:

- **New Capability:** 100% of the application functionality—from menu navigation to lesson completion—is accessible via a **webcam-based eye-tracking system**.
- **Dwell-to-Click:** It utilizes a "**dwell-to-click**" mechanism (gazing at a target to select it), eliminating the need for fine motor control.

3. A Dynamic "AI Adaptive Lesson Engine"

- **Real-Time Generation:** The AI generates micro-lessons on the fly based on user performance, rather than pulling from a fixed database.
- **Context-Aware Adaptation:** The AI reacts to specific behaviors. For example, if a user answers correctly but very quickly, the AI infers

"boredom" and increases the novelty of the next question. Conversely, if a user struggles, it breaks the problem down into simpler parts.

4. Integrated "Focus-Aware" Technology

The system actively monitors user attention in a way traditional apps do not.

- **Focus Tracking:** It passively tracks "gaze-wander" (looking away from the screen) to detect distraction
- **Non-Punitive Nudges:** Instead of penalizing the student, the system uses this data to offer gentle, "intrinsic" nudges (like a breathing highlight) to bring their focus back to the task.

Project Objectives

- 1. Personalization (AI):** To deliver a 1:1, AI-driven learning path that continuously adapts the content's difficulty and style to the individual user.
- 2. Accessibility (CV):** To enable 100% of the application's functionality to be controlled via a software-only, webcam-based eye-tracking system, eliminating the need for physical input.
- 3. Engagement (UX):** To maximize on-task focus and lesson completion rates using gamified mechanics and an interface specifically designed for ADHD (attention deficit hyperactivity disorder).
- 4. Autonomy (System):** To create a "closed-loop" system where the student can independently manage their learning experience—from calibration to lesson completion—without requiring external help.

Related Work

The landscape of accessible educational technology is currently bifurcated, with distinct solutions addressing either physical motor impairments or cognitive challenges (ADHD), but rarely both simultaneously. EduFocus Adaptive emerges as a novel integration of these two historically separate domains.

1. Eye-Tracking & Physical Accessibility

Current solutions for students with severe motor impairments (e.g., Cerebral Palsy, Spinal Muscular Atrophy) predominantly rely on hardware-intensive assistive technologies.

- **Hardware-Based Solutions:** Market leaders like Tobii Dynavox provide sophisticated eye-tracking hardware (e.g., I-Series) that enables communication and general computer access. While effective, these systems often require expensive proprietary hardware and are designed primarily for "communication" (AAC) or general OS navigation rather than dedicated, pedagogically structured learning.
- **The EduFocus Distinction:** Unlike these general-purpose hardware tools, EduFocus Adaptive utilizes a **software-only, webcam-based approach**. By leveraging on-device computer vision (TensorFlow Lite/MediaPipe), it democratizes access, allowing any device with a standard webcam to function as a high-precision input method without specialized peripherals

2. Adaptive Learning & Gamification for ADHD

For students with ADHD, the "Cognitive Barrier" is typically addressed through gamification and adaptive algorithms designed to sustain attention.

- **Existing Platforms:** Applications like DreamBox and MotionMath utilize adaptive learning to adjust content difficulty based on student performance. Similarly, tools like "Read&Write" offer text-to-speech support to aid focus.

However, these platforms almost exclusively rely on traditional input modalities—mouse, keyboard, or touch.

- **The EduFocus Distinction:** While standard apps use static content libraries, EduFocus employs a **Generative AI Engine (Gemini API)** to create micro-lessons in real-time. Furthermore, it introduces "Focus-Aware" technology that passively monitors gaze-wander to detect distraction, offering non-punitive "nudges" rather than simple error correction.

3. The Gap of Intersectionality

A critical void exists at the intersection of these two fields.

- **The "Double Burden":** Research highlights that students often face a "double burden" where accessible tools are not engaging for neurodivergent minds, and engaging tools are physically inaccessible. For instance, a gamified math app might be cognitively perfect for a child with ADHD but unusable if that same child cannot use a touchscreen due to motor impairments.
- **Holistic Integration:** EduFocus Adaptive addresses this by creating a **"Dual-Engine" system**. It is one of the few proposed frameworks to pair **gaze-based interaction** (for physical access) directly with **generative AI adaptation** (for cognitive engagement) in a single, self-contained loop. This effectively removes the need for external physical inputs while simultaneously managing the user's cognitive load.

System Features

The system is built to function as a self-contained, autonomous learning loop for students with ADHD and motor impairments. Its features are categorized below by their primary function.

1. Core Pedagogical Features (The AI Engine)

These features are powered by the AI Adaptive Lesson Engine, which generates content in real-time rather than relying on static databases.

- **Real-Time Lesson Generation:** The system generates interactive micro-lessons (e.g., multiple-choice questions, matching tasks) on the fly using Generative AI (Gemini API).
- **Dynamic Difficulty Adjustment:** The AI adapts the next lesson based on immediate user performance. If a user masters a topic (3+ correct answers), difficulty increases; if they struggle (2+ incorrect), it breaks the problem down into simpler foundational steps.
- **Text-to-Speech Integration:** All generated text and instructions are automatically converted to speech to support students with reading difficulties.
- **Novelty Injection:** If a user answers correctly but too quickly (indicating boredom), the system automatically changes the question style (e.g., from multiple choice to fill-in-the-blank) to maintain engagement.

2. Interaction & Control Features (The Vision Engine)

These features constitute the Eye-Tracking Control System, enabling a "zero-touch" interface.

- **Webcam-Based Gaze Tracking:** The system utilizes a standard webcam to track eye movements, rendering a "gaze cursor" on-screen that mirrors the user's focus with low latency (<150ms).
- **Dwell-to-Click Selection:** Users interact with elements by holding their gaze on a target for a configurable duration (default: 1.5 seconds). A visual ring indicator shows progress toward the "click".
- **Gaze-Based Scrolling:** Navigation through menus or long content is achieved by looking at the top or bottom 10% of the screen.
- **Calibration Wizard:** A one-time, visually guided setup process creates a custom eye-tracking profile for the user by having them look at a series of 5-9 targets.

3. Engagement & Focus Features

These features are specifically designed to support neurodivergent students (e.g., ADHD) by maintaining attention and reducing frustration.

- **Gamified Micro-Lessons:** Content is structured into short, 3-5 minute "missions" to align with shorter attention spans.
- **Passive Focus Tracking:** The system monitors "gaze-wander" (looking away from the screen). If the user is off-task for >10 seconds, it triggers a "focus aid".
- **Non-Punitive Nudges:** Instead of punishing distraction, the system uses subtle visual cues (e.g., a breathing highlight) or quiet audio prompts ("Still working on this?") to gently re-engage the user.
- **Intrinsic Feedback:** Positive feedback is immediate and visual (e.g., a puzzle piece fitting), while errors are met with gentle guidance ("Good try, let's look again") rather than harsh alerts.

4. Privacy & Technical Features

These non-functional features ensure the system is safe and deployable on standard hardware.

- **On-Device Biometric Processing:** All camera video processing and eye-tracking analysis are performed 100% locally on the device using TensorFlow Lite; no video frames or face mesh data are ever sent to a server.
- **Low-Resource Requirement:** The application is optimized to run efficiently ($\leq 30\%$ CPU load) on mid-range hardware with a standard 720p webcam.

CHAPTER TWO

2 SYSTEM ANALYSIS AND DESIGN

INTRODUCTION

The rapid expansion of digital learning technologies has transformed modern education, yet a significant group of learners remains excluded—students who face both cognitive challenges such as ADHD and severe physical motor impairments. As highlighted in the EduFocus project documentation, traditional learning systems depend heavily on static content and physical interaction through mouse, keyboard, or touchscreens, creating substantial barriers for these students. At the same time, most adaptive or gamified learning tools focus solely on cognitive engagement without considering physical accessibility needs. This fragmentation leaves a critical gap where no holistic solution exists to accommodate learners who experience both cognitive and physical limitations simultaneously.

EduFocus Adaptive was conceived to directly address this “double barrier” by applying structured system analysis and thoughtful design principles. The Product Requirements Document describes the system as a next-generation educational platform built on a dual-engine architecture: an **AI Adaptive Lesson Engine** that dynamically generates personalized micro-lessons, and a **Computer Vision–based Eye-Tracking Control System** that enables complete, hands-free navigation through gaze interaction.

This approach aligns with the project’s guiding philosophy of *radical inclusion*—the belief that technology must adapt to the user, not the other way around.

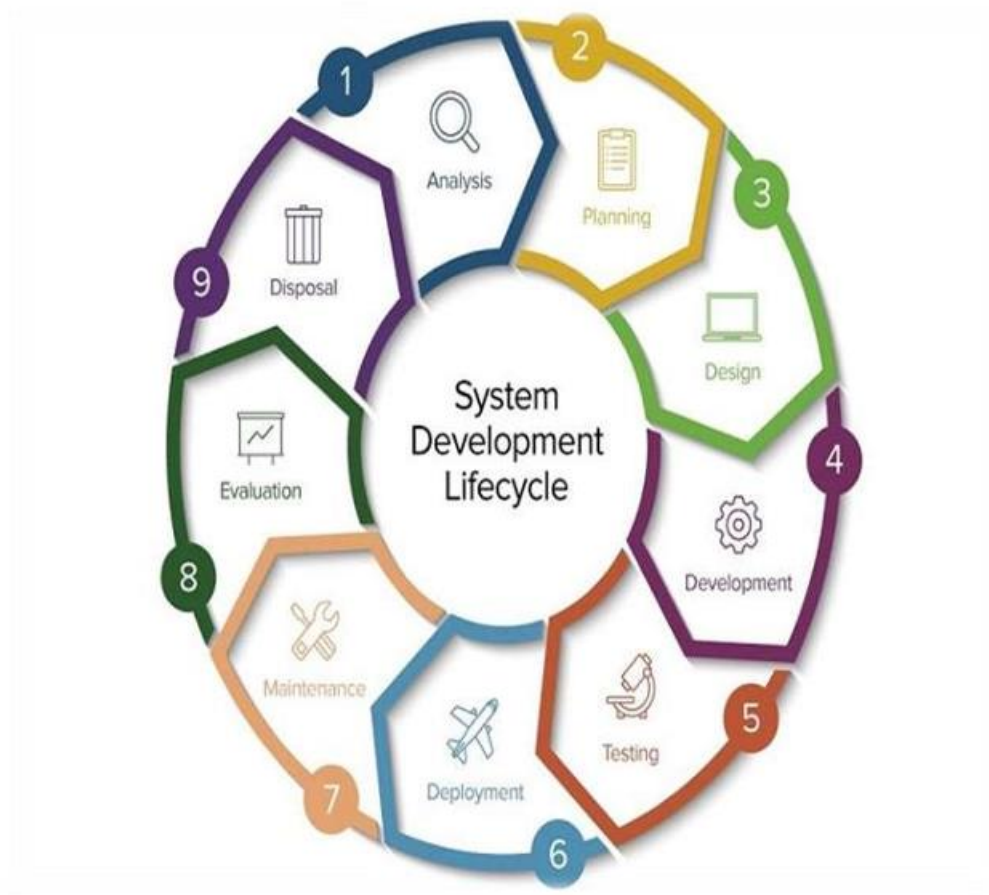
From a systems analysis perspective, EduFocus Adaptive aims to solve a clearly defined problem through a cohesive set of functional and non-functional requirements. The platform enables fully autonomous learning, providing real-time adaptation to the learner's performance and attention while ensuring complete accessibility through webcam-based interaction. System design decisions—from the use of on-device processing for privacy, to the simplified user flow and gaze-based controls—are driven by the unique needs of the target user persona: a child with ADHD and motor impairments who requires both cognitive support and a zero-touch interface to participate effectively in digital learning environments.

This introduction sets the foundation for a detailed System Analysis and Design document. It clarifies the problem space, defines the system's purpose, and establishes the need for a dual-engine architecture capable of delivering personalized, autonomous, and fully accessible education.

SYSTEM ANALYSIS

Planning

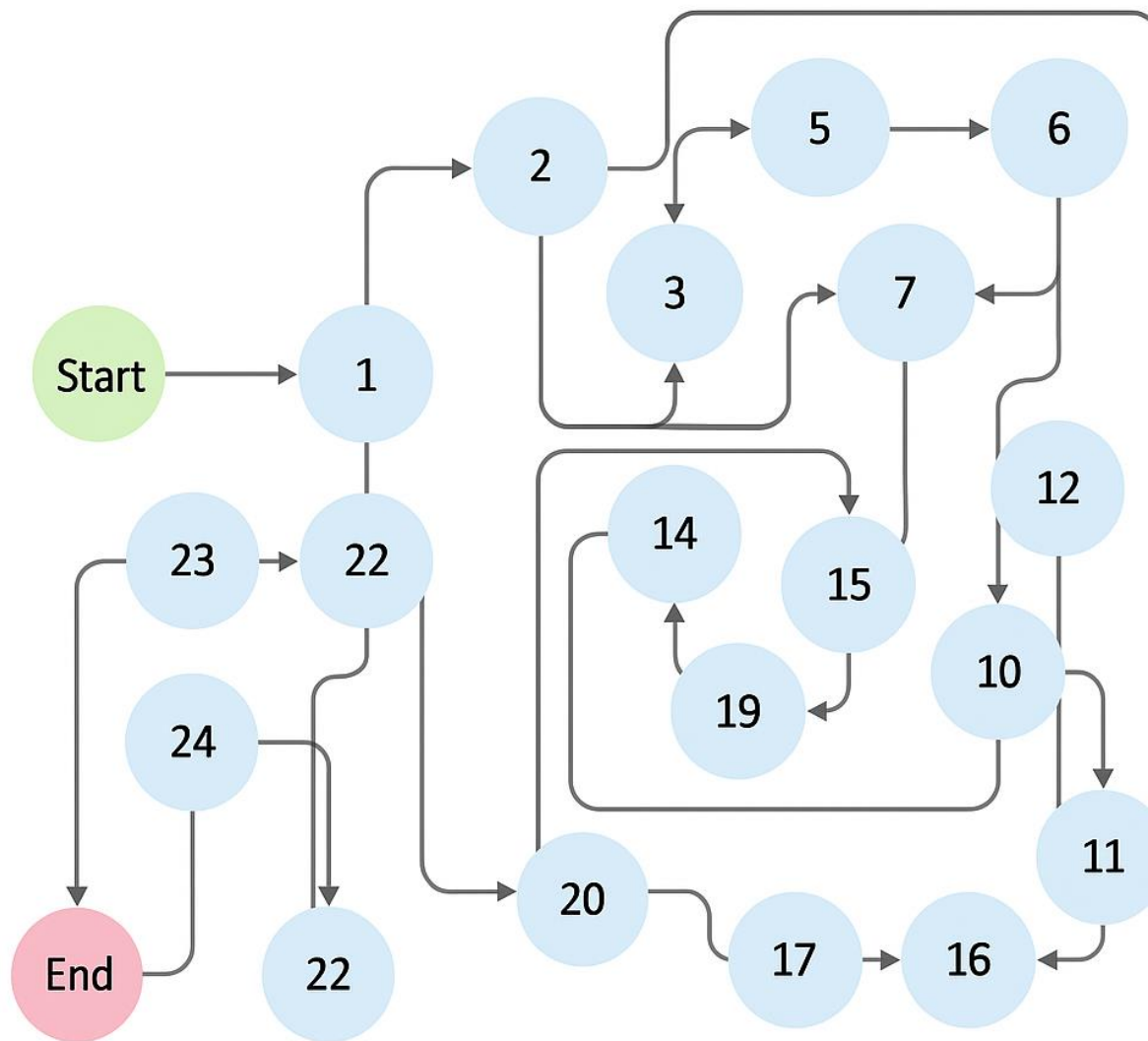
System Development Life Cycle



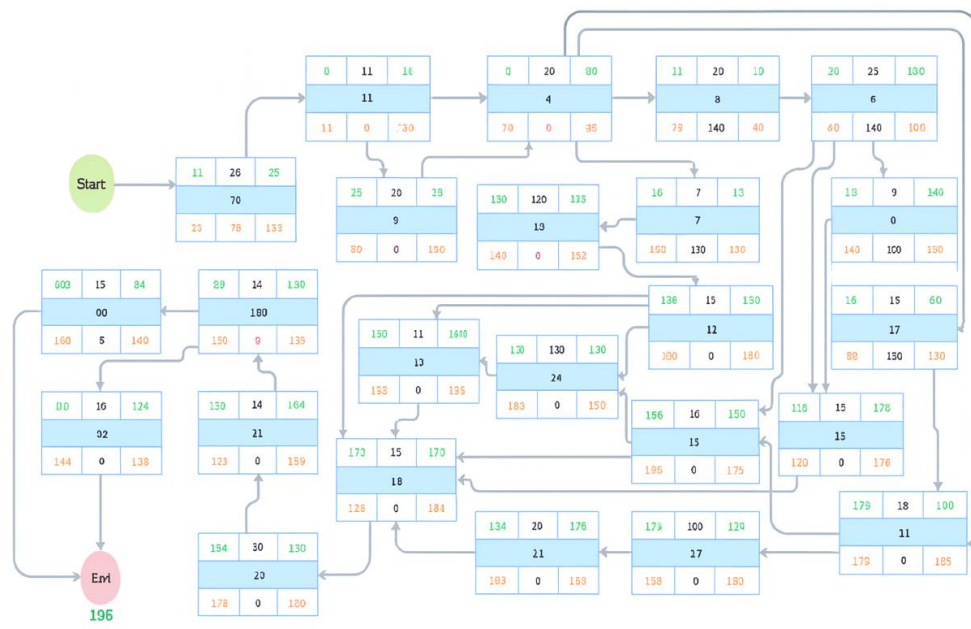
Task Scheduling Table

Index	Task	Predecessors	Start Date	End Date	Duration
1	Project Idea & Vision		8/25/2025	9/05/2025	11d
2	Requirements Gathering	1	9/06/2025	9/21/2025	15d
3	Feasibility & Planning	2, 3	9/22/2025	10/04/2025	12d
4	System Analysis	4, 5	10/05/2025	11/04/2025	30d
5	System Design (Architecture)	4	11/05/2025	11/19/2025	15d
6	UVUX Design	5, 5	11/20/2025	12/10/2025	20d
7	Eye-Tracking Engine Development	5	12/11/2025	1/20/2026	40d
8	AI Lesson Engine Development	5	12/11/2025	1/25/2026	45d
9	Front-End Development	6	1/02/2026	2/03/2026	32d
10	Backend API Development	7, 8	2/04/2026	2/18/2026	14d
11	Integration: AI + CV Engi-	7, 8, 1	2/19/2026	3/04/2026	14d
12	Eye-Tracking Testing	7	1/26/2026	1/30/2026	4d
13	AI Engine Testing	8	1/26/2026	2/05/2026	10d
14	Front-End Testing	9, 9	2/04/2026	2/10/2026	6d
15	Improve Eye-Tracking Accuracy	12, 13	2/11/2026	2/25/2026	14d
16	Improve AI Adaptation Model	13, 13	2/11/2026	2/26/2026	15d
17	Connect Front-End to API	10, 14, 14	2/27/2026	3/05/2026	7d
18	Combined System Testing	11, 15, 16, 17	3/06/2026	3/16/2026	10d
19	Deploy CV + AI Modules	11, 18, 18	3/17/2026	3/21/2026	4d
20	System Unit Testing	19	3/22/2026	3/29/2026	7d
21	Svstem Integration Testing	20	3/30/2026	4/07/2026	8d

Network diagram



PERT



Gantt Chart



System Requirements

Functional Requirements

The functional requirements of the EduFocus Adaptive system define the core operations and behaviors necessary to achieve its objectives of accessibility, personalization, and autonomous learning. The system should provide a fully hands-free interaction model using a webcam-based eye-tracking mechanism that enables users to navigate the interface, select options, and scroll content through gaze and dwell-to-click interaction. It shall generate adaptive educational content in real time using an AI-driven lesson engine that adjusts difficulty, presentation style, and feedback based on the learner's performance and focus level. The system shall support a calibration process to accurately map user eye movements, offer text-to-speech functionality for all instructional content, and provide immediate, non-punitive feedback for both correct and incorrect responses. Additionally, the system shall record user performance and focus metrics to continuously refine lesson adaptation, ensuring an engaging, accessible, and independent learning experience for students with cognitive and motor impairments.

User & Calibration

- **FR-5.1:** The system shall support a single "Student" user profile, stored locally on the device.
- **FR-5.2:** On first launch, the system must initiate the Eye-Tracking Calibration Wizard.
- **FR-5.3:** The Calibration Wizard shall consist of the user "looking at" a series of 5-9 on-screen targets.
- **FR-5.4:** The user must be able to re-initiate calibration at any time from the main menu.

Eye-Tracking & Interaction

- **FR-5.5:** The system shall render a "gaze cursor" on-screen that mirrors the user's eye movement.
- **FR-5.6:** Selection shall be performed by "Dwell-to-Click": holding the gaze cursor on an interactive element for a configurable duration (default: 1.5 seconds).
- **FR-5.7:** A visual indicator (e.g., a closing ring) must show dwell-to-click progress.
- **FR-5.8:** The system shall support gaze-based scrolling (e.g., looking at the top/bottom 10% of the screen).

AI & Learning

FR-5.9: The system shall present a simple "Subject" selection menu (e.g., Math, Reading, Science).

FR-5.10: Upon subject selection, the AI Engine shall generate a single, interactive micro-lesson (e.g., a multiple-choice question, a "match the pair" task).

FR-5.11: All lesson content (text, instructions) must be automatically converted to speech (Text-to-Speech).

FR-5.12: The AI must adapt to the next lesson based on the user's interaction with the current one (see Section 9: AI Design System).

FR-5.13: The system must provide immediate, positive, and non-distracting audio-visual feedback for correct answers.

FR-5.14: For incorrect answers, the system shall provide a gentle, non-punitive "nudge" (e.g., "Good try! Let's look at that again...") and a simplified hint.

Non-Functional Requirements

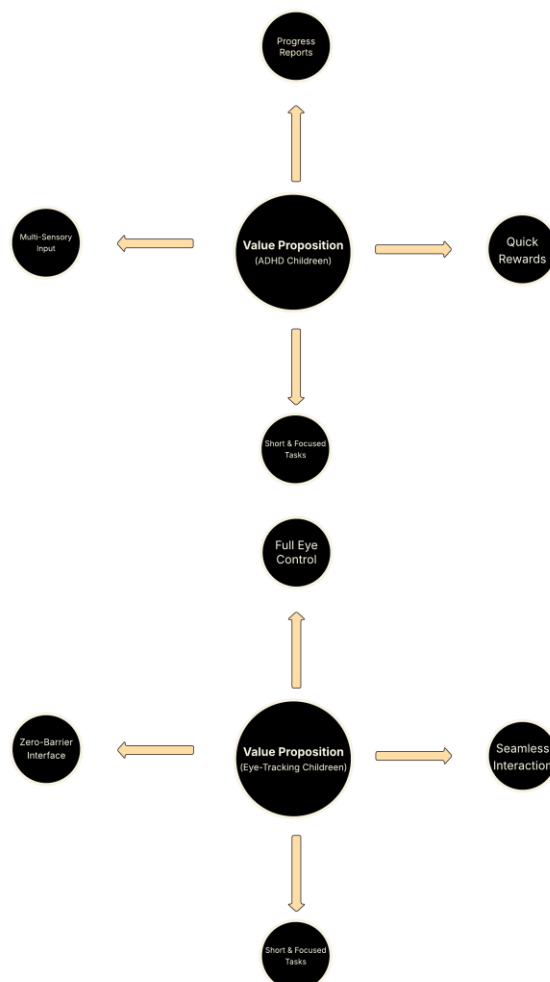
The non-functional requirements of the EduFocus Adaptive system define the quality attributes that ensure reliability, usability, performance, and security. The system shall provide real-time responsiveness, with eye-tracking interactions exhibiting minimal latency to ensure smooth and natural user control. It should operate efficiently on mid-range hardware using a standard webcam, maintaining acceptable CPU and memory usage. To protect user privacy, all biometric data, including eye-tracking and camera input, shall be processed locally on the device, with no raw video or sensitive data transmitted externally. The user interface shall adhere to accessibility and usability standards by maintaining a low-clutter design, high contrast visuals, large interactive elements, and clear typography suitable for neurodivergent users. Additionally, the system shall be robust against variations in lighting conditions and user appearance, and it shall ensure system availability, fault tolerance, and data integrity to support a safe, reliable, and frustration-free learning environment.

- NFR-6.1: Performance
 - Eye-Tracking: Gaze-to-cursor latency must be < 150ms to feel responsive.
 - AI Generation: A new micro-lesson (question + options) must be generated and displayed in < 4 seconds.
 - System: The application must run efficiently ($\leq 30\%$ CPU load) on mid-range hardware with a standard 720p/1080p webcam.
- NFR-6.2: Privacy & Security
 - CRITICAL: All camera video processing and eye-tracking analysis must be performed 100% on-device. No video frames or biometric data (e.g., face mesh) shall be sent to any server.

- Only anonymized, non-biometric data (e.g., lesson prompts, performance statistics) may be sent to the backend for AI generation and storage.
- NFR-6.3: Accessibility (UI/UX)
 - The app must adhere to a "low-clutter," high-contrast design.
 - All interactive elements ("gaze targets") must have a minimum size of 100x100dp.
 - All text must be rendered in a clear, sans-serif, or dyslexic-friendly font.
 - No core functionality shall be "hidden" in complex menus.
- NFR-6.4: Robustness
 - The eye-tracking system must be robust to variations in lighting (e.g., daytime indoor) and user accessories (e.g., glasses).

Business Model

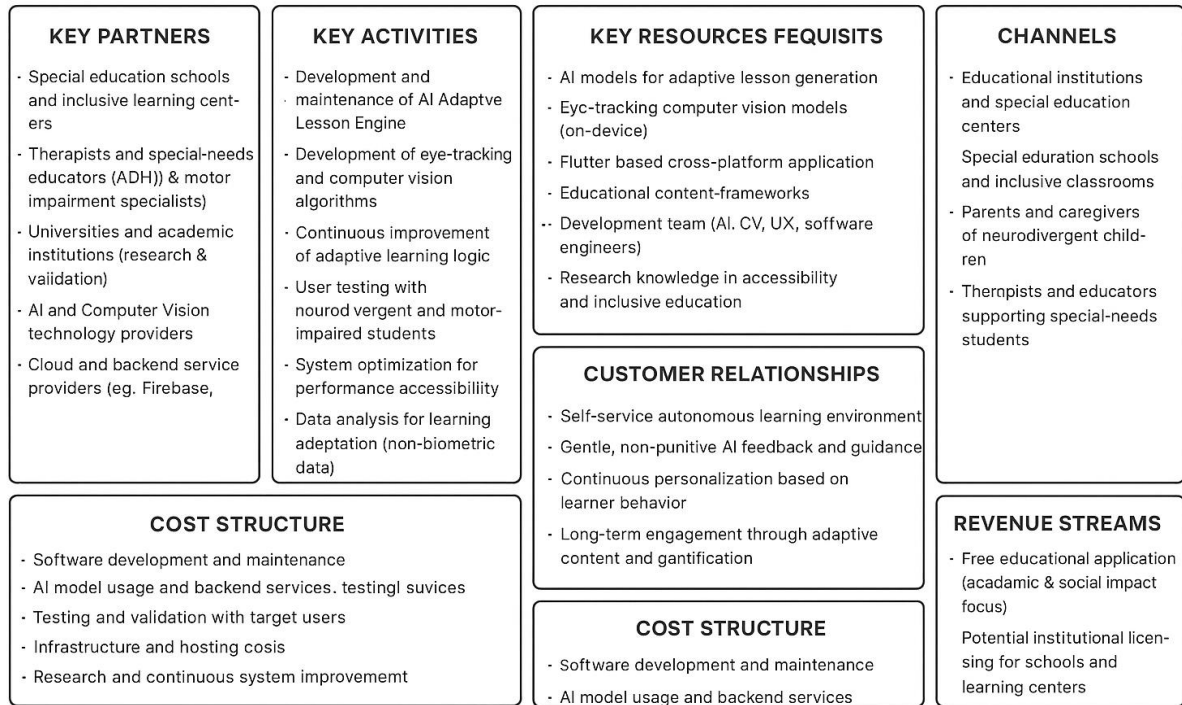
Value Proposition



Business Model Canvas

BUSINESS MODEL CANVAS

EduFocus Adaptive



System Design

Stack holders

1. Child Actors

(Includes Children with ADHD and Physically Disabled Children) – The end beneficiary and primary consumer of the learning content –:

- **Authentication:**
 - Login/Logout: Ability to securely enter and exit the system after a session.
- **Content Access:**
 - View Lessons: Access and browse the available educational curriculum organized by academic level.
- **Content Interaction:**
 - ADHD Children: Capability to Play Learning Games and view correct answers (View Answers) during the interactive sessions.
 - Disabled Children: Interact with the lessons through the alternative Eye-Tracking interface to access the core content.
- **Motivation & Monitoring:**
 - Receive immediate rewards (Get Rewards) upon successful completion.
 - Track personal progress (Track Progress) within the curriculum.
 - Receive notifications (Receive Notifications).

2. Parent

(Father, Mother) – The Account Administrator and Performance Monitor –:

- **Account Management:**
 - Capability to Register to create a main account and then manage the child's profile through updating (Update Child Profile) or deleting (Delete Child Profile) it.
- **System Configuration:**
 - Ability to Select the appropriate interaction Mode for the child (ADHD Game Mode or Eye-control Mode).
- **Monitoring & Reporting:**

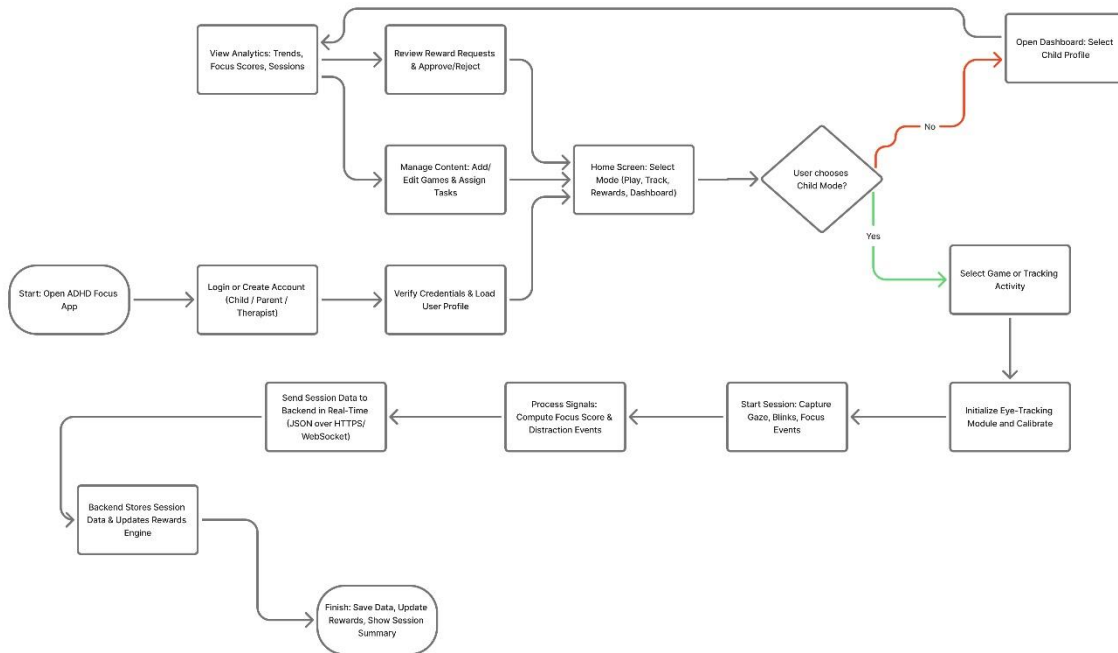
- Monitor Child Progress directly within the Parent dashboard.
- Generate detailed, exportable reports (Get Reports) concerning the child's academic performance and attention metrics.
- **System Exit:**
 - Capability to securely Logout from their account after managing settings.

3. Admin

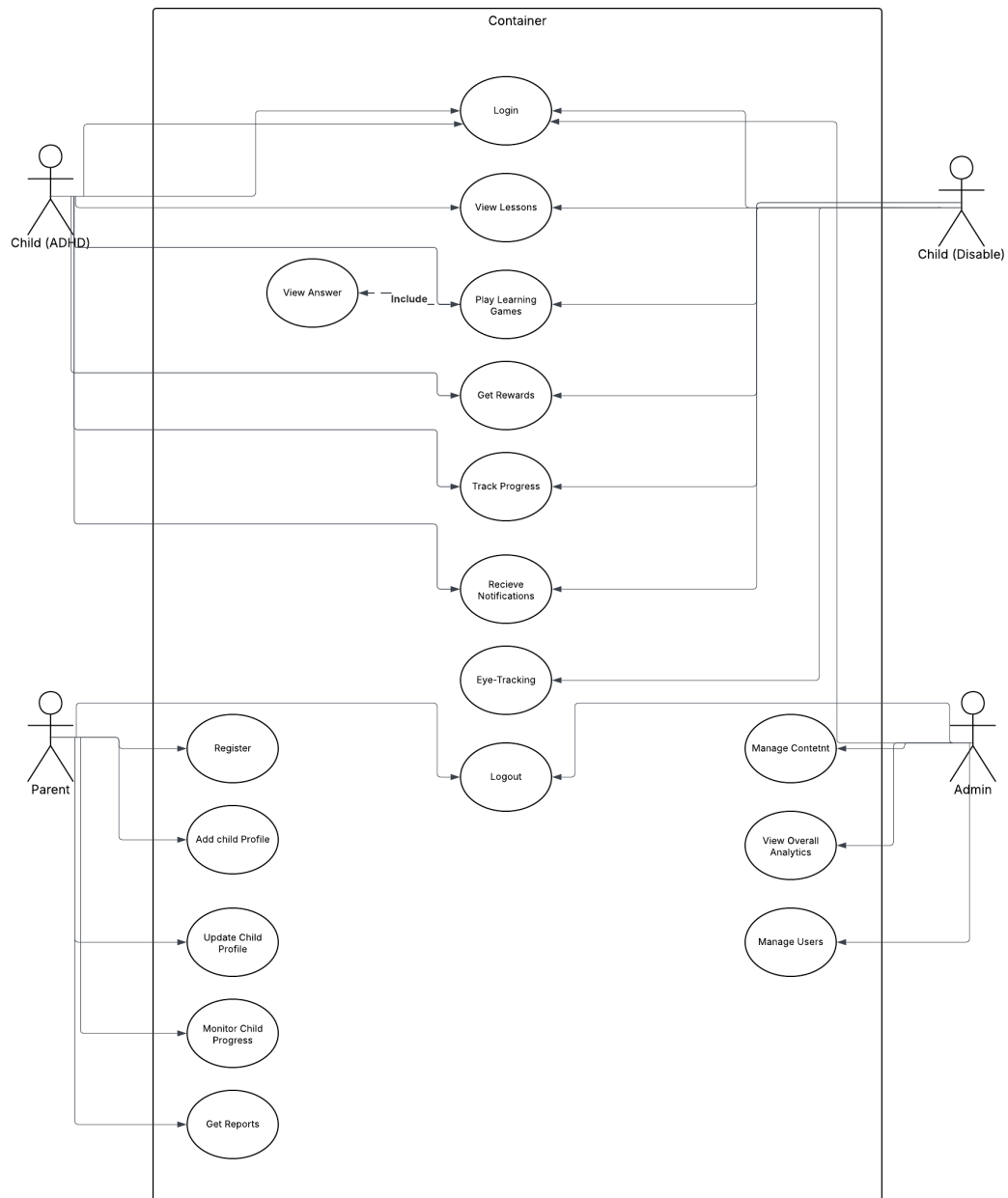
(System Owner and General Supervisor) – The individual responsible for system maintenance, content, and users –:

- Content Control:
 - Full control to Manage Content, including adding, modifying, or deleting educational curricula and related learning activities/games.
- User Management:
 - Ability to Manage Users, including controlling parent accounts and resolving login or profile issues.
- System Analytics:
 - Access comprehensive analytics (View Overall Analytics) and statistics regarding system usage, feature popularity, and overall performance for strategic decision-making.
- System Exit:
 - Capability to securely Logout from the administrative dashboard.

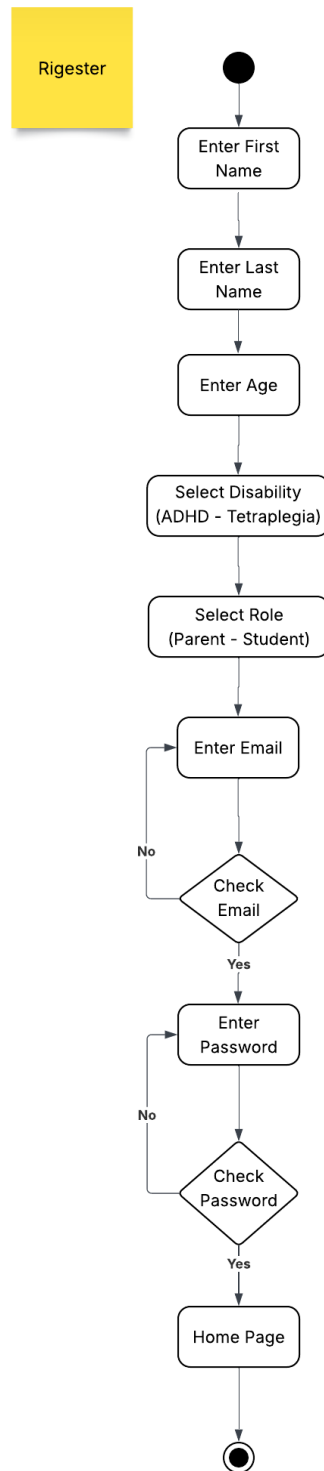
Block Diagram



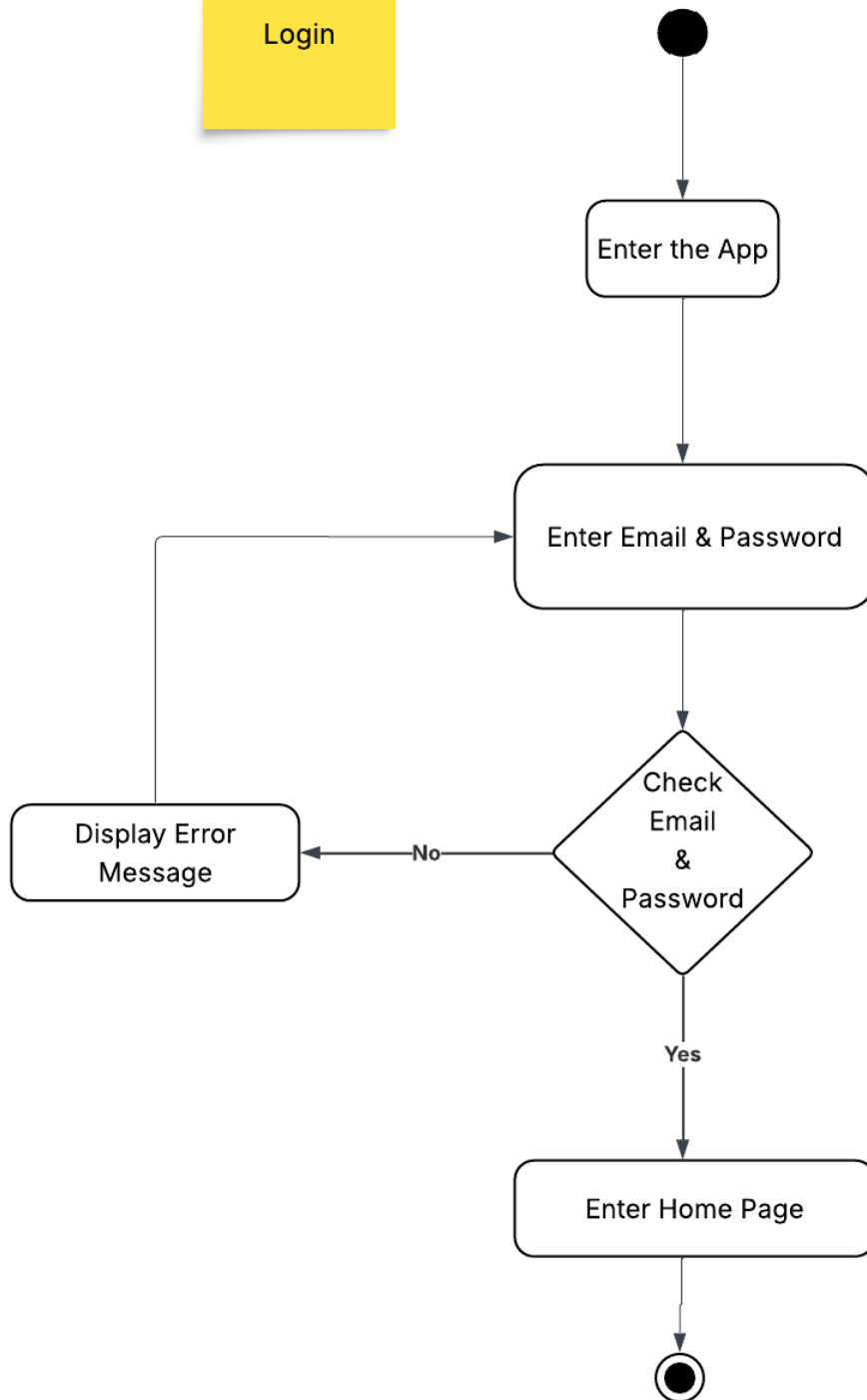
Use Case



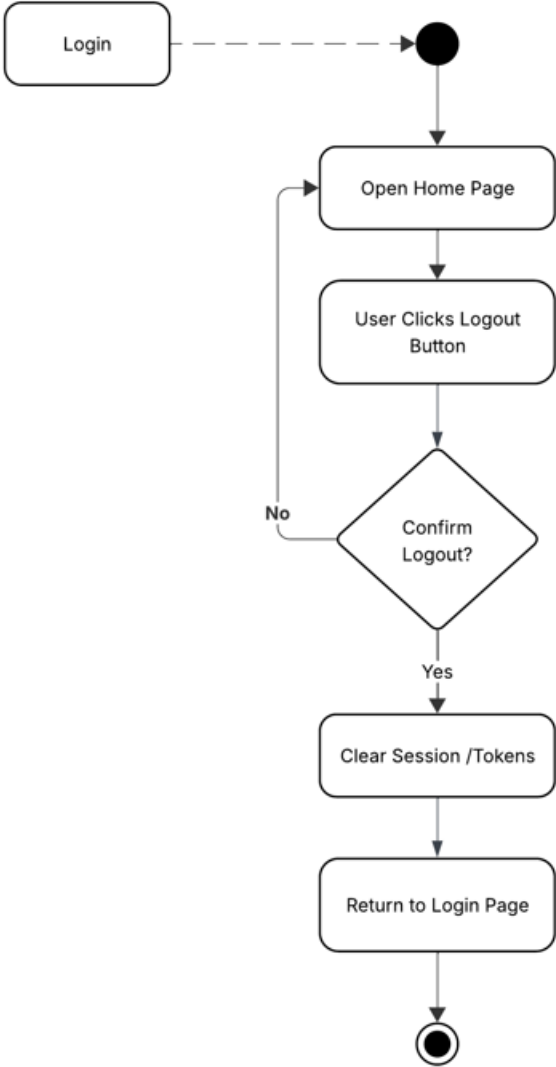
Activity Diagram



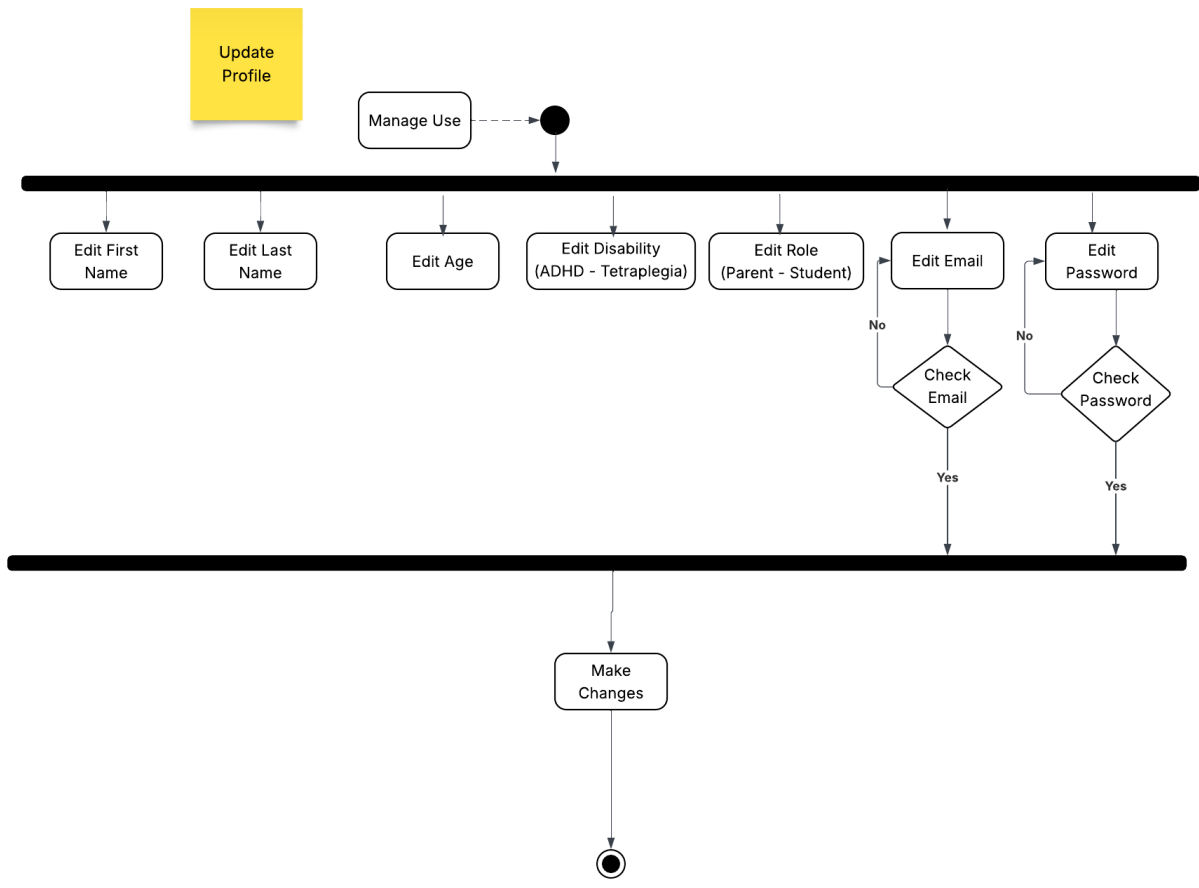
Login



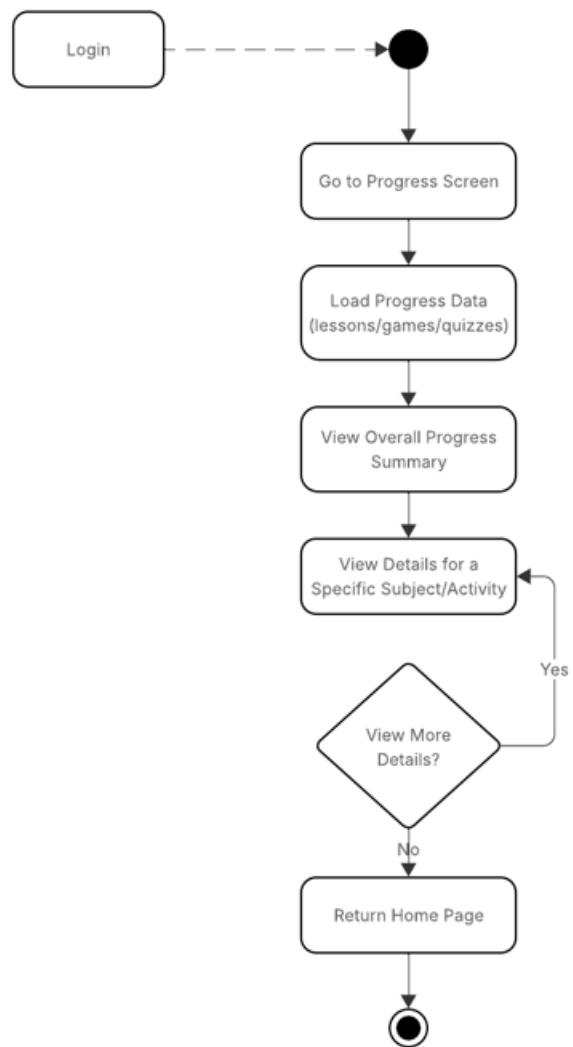
Logout



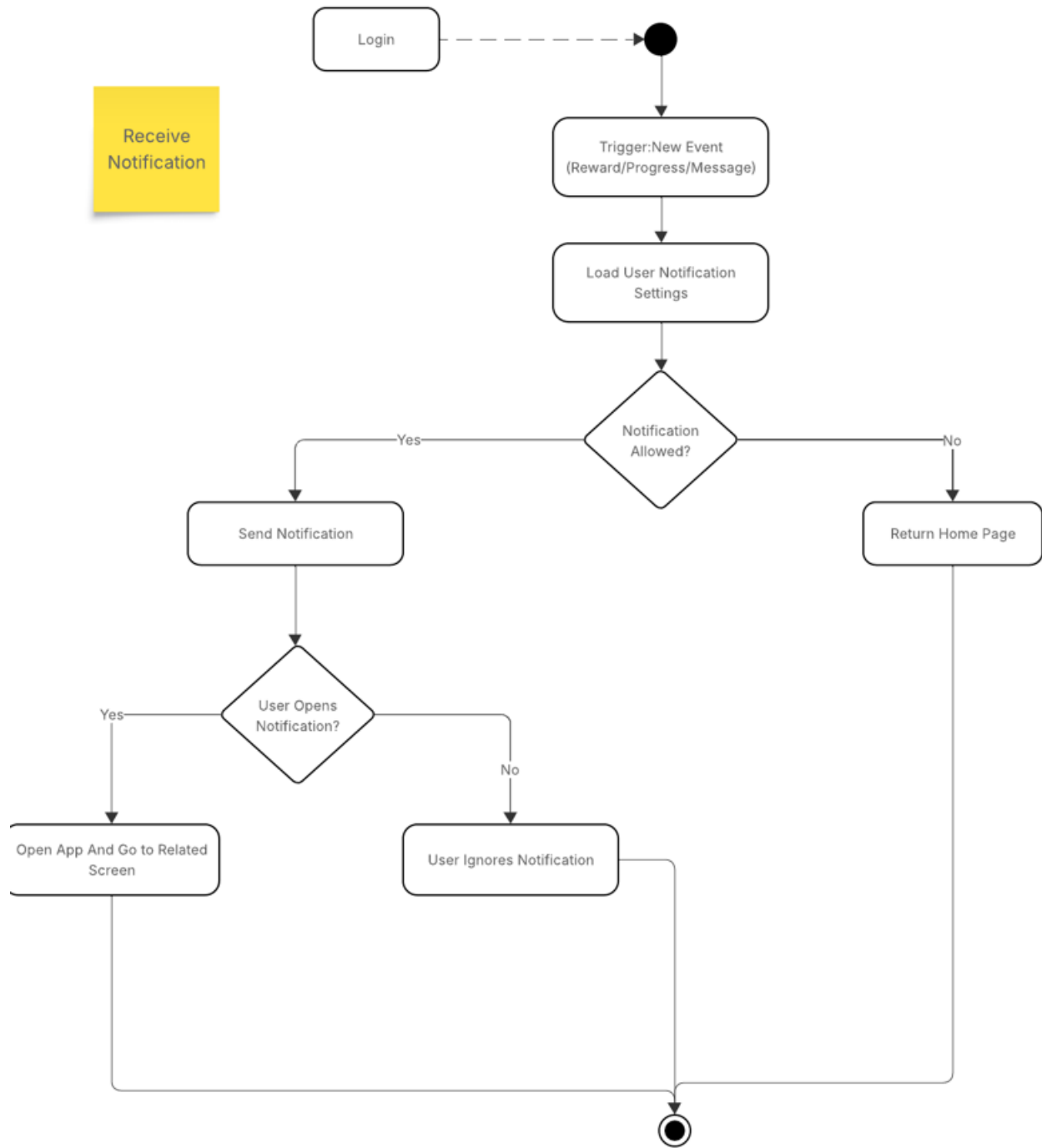


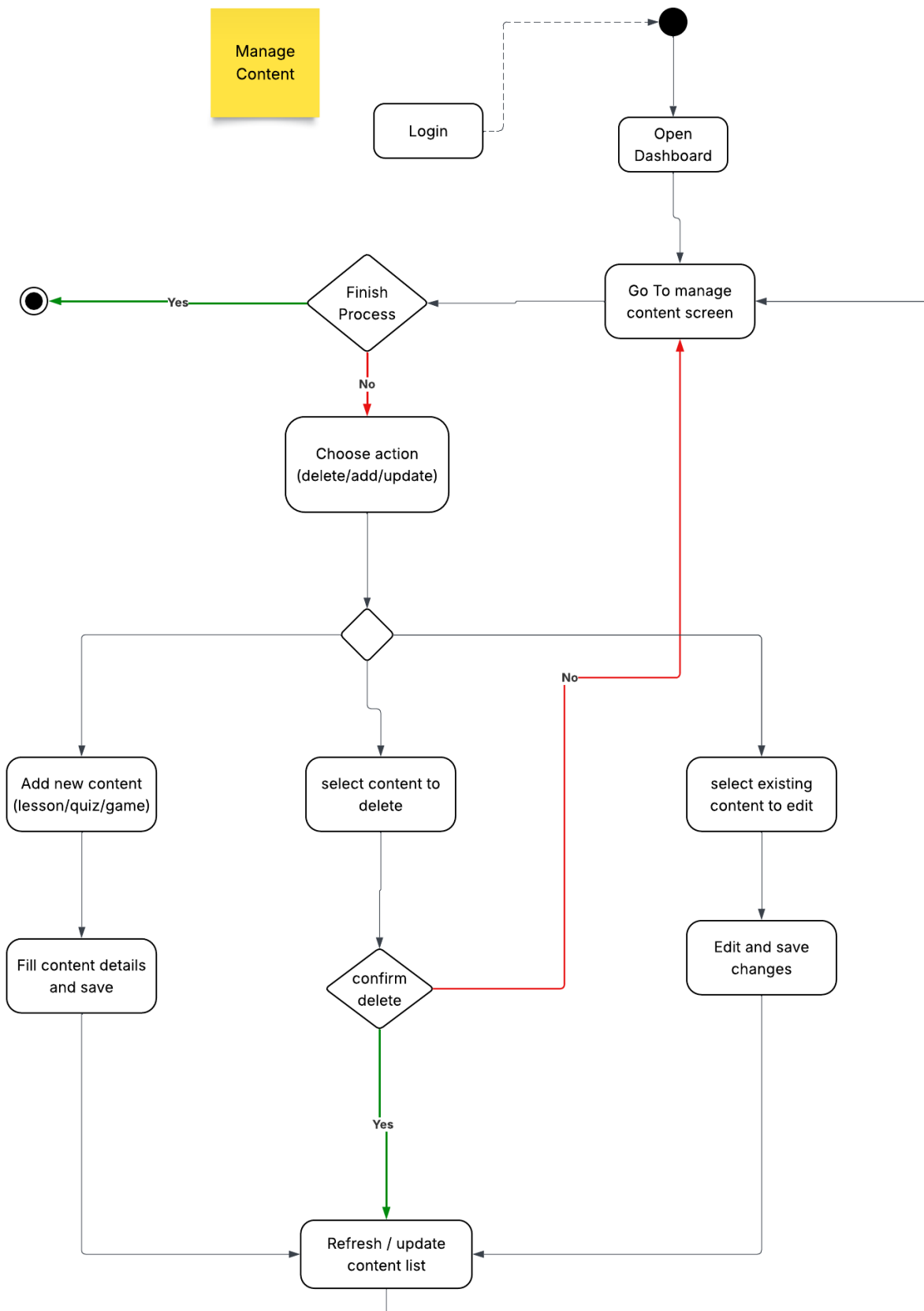


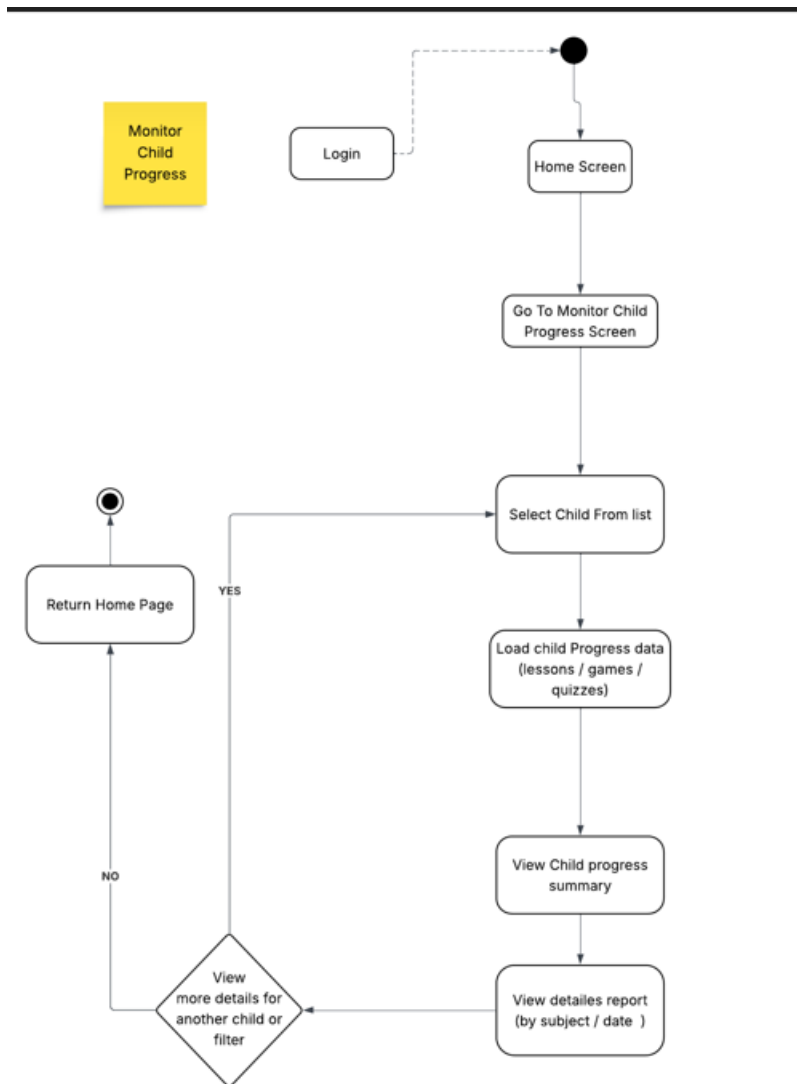
Track Progress

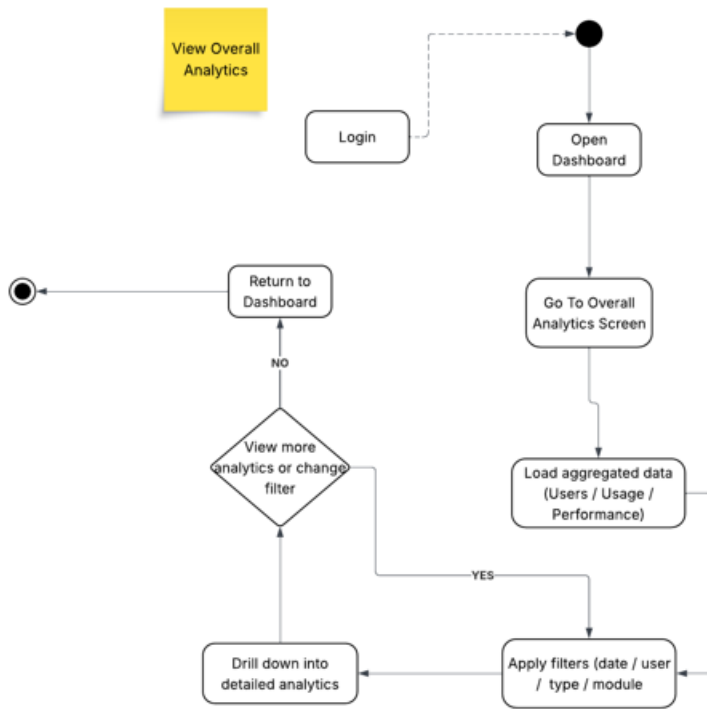


Receive
Notification

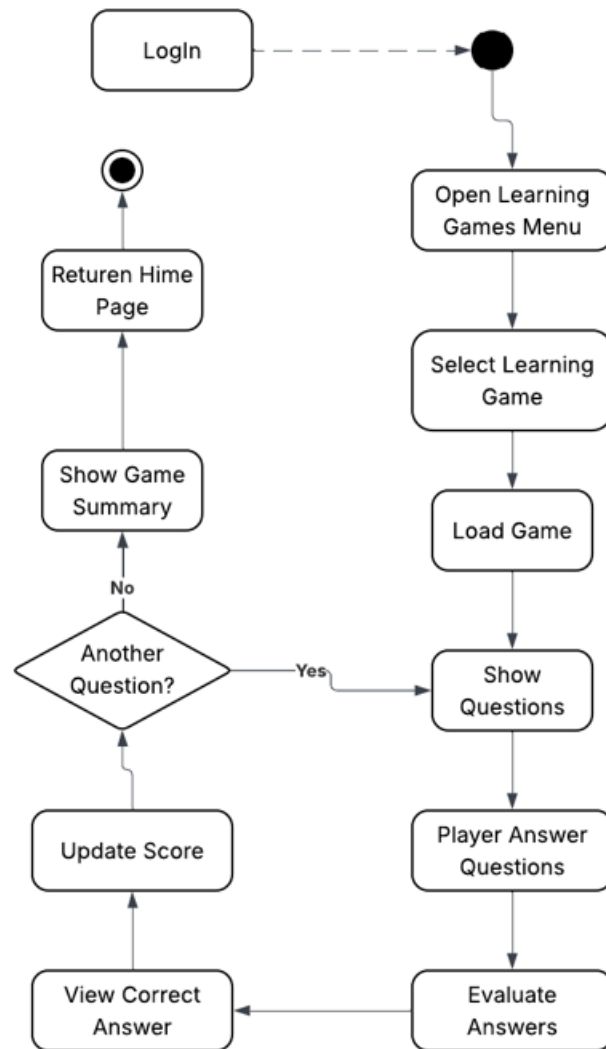




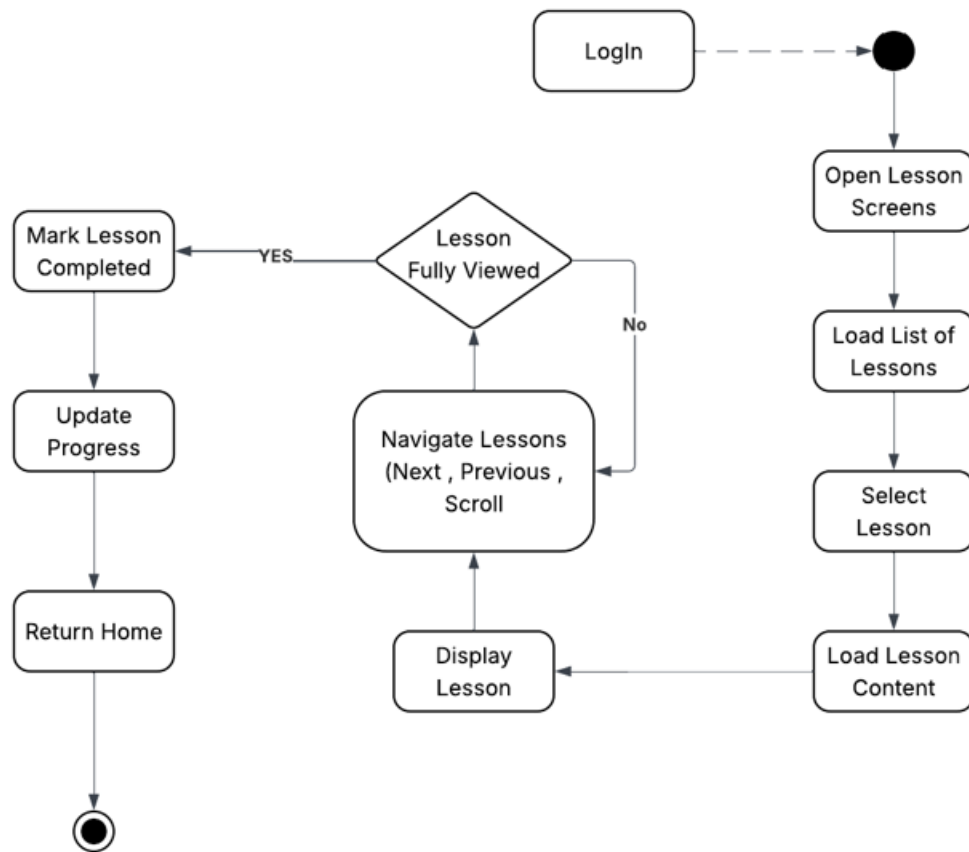




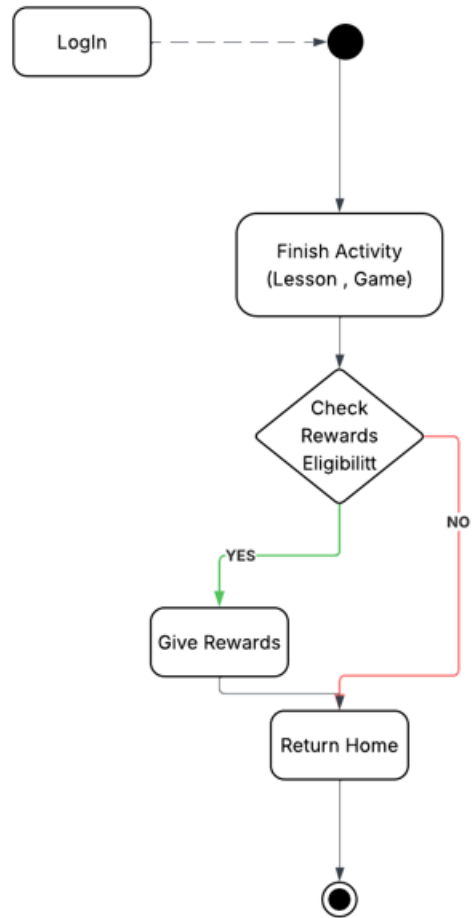
Playing Games



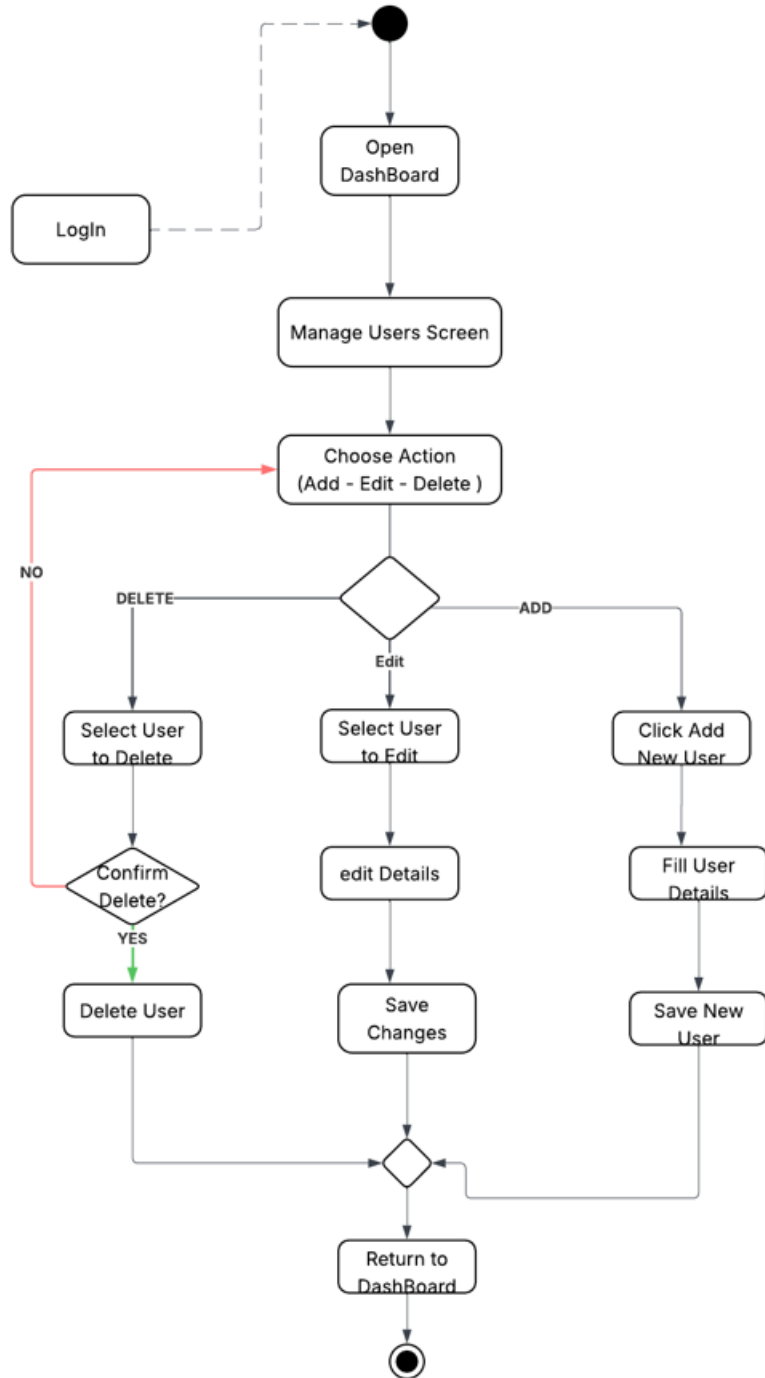
View Lessons



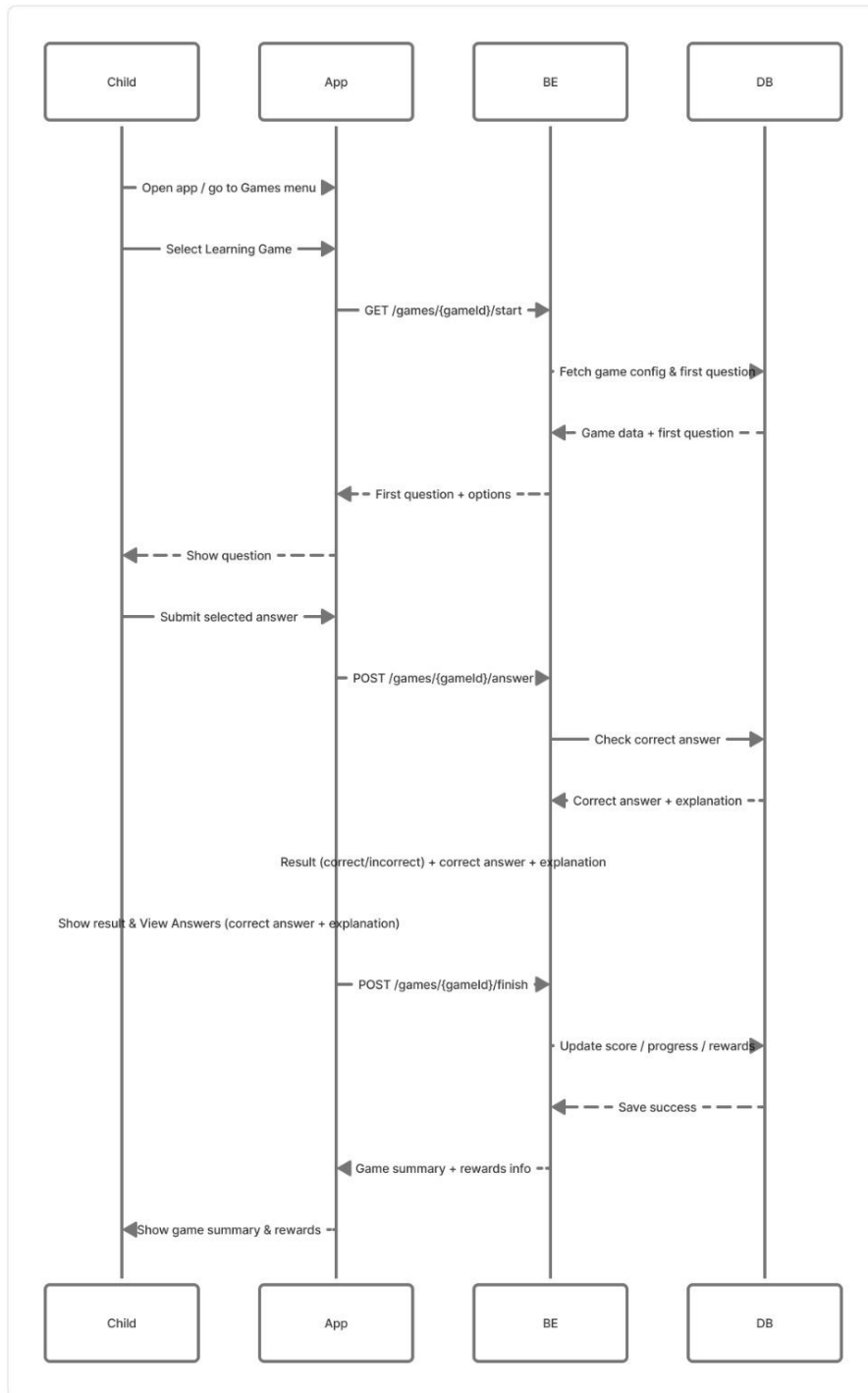
Get Rewards

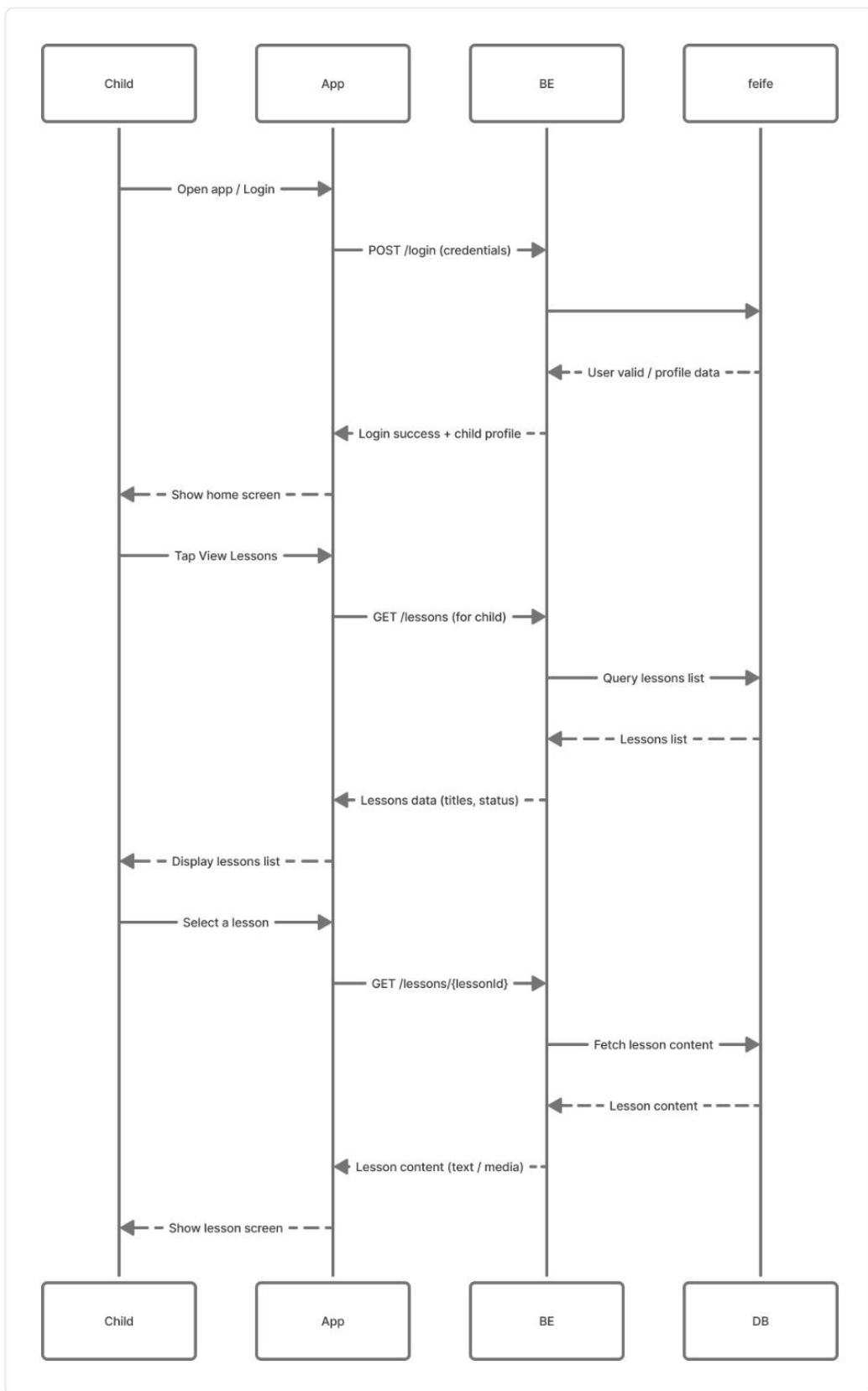


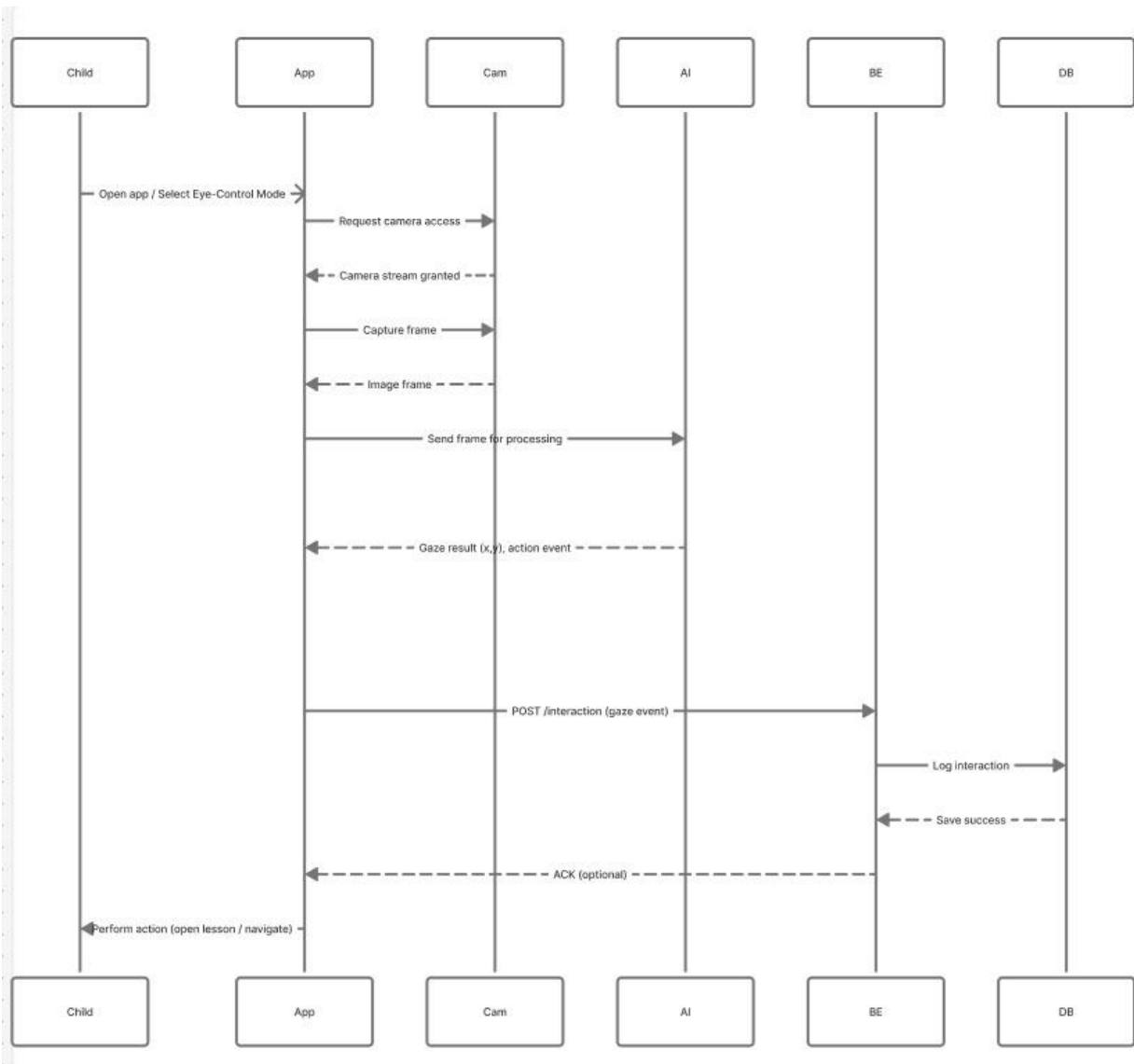
Mange Use

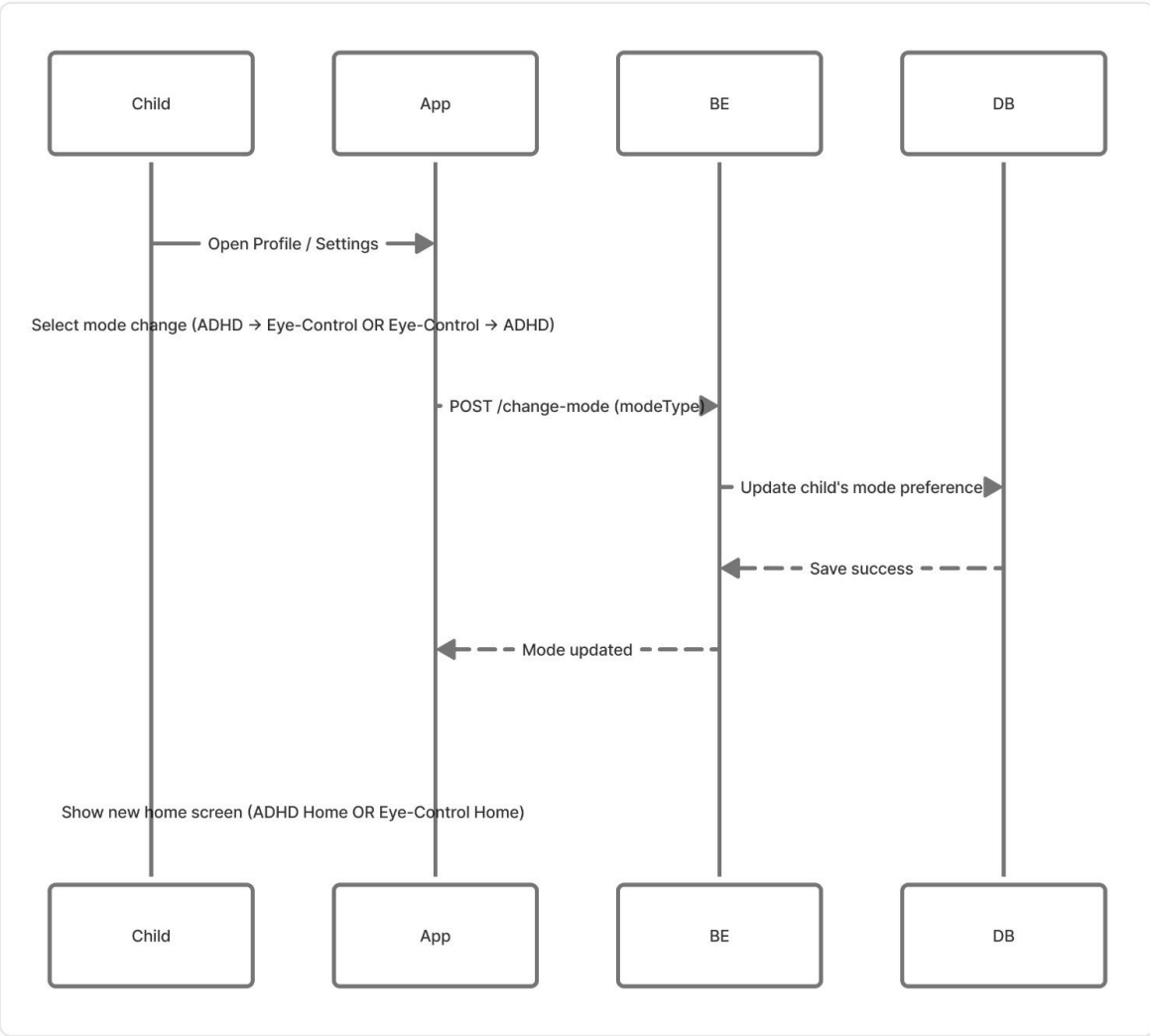


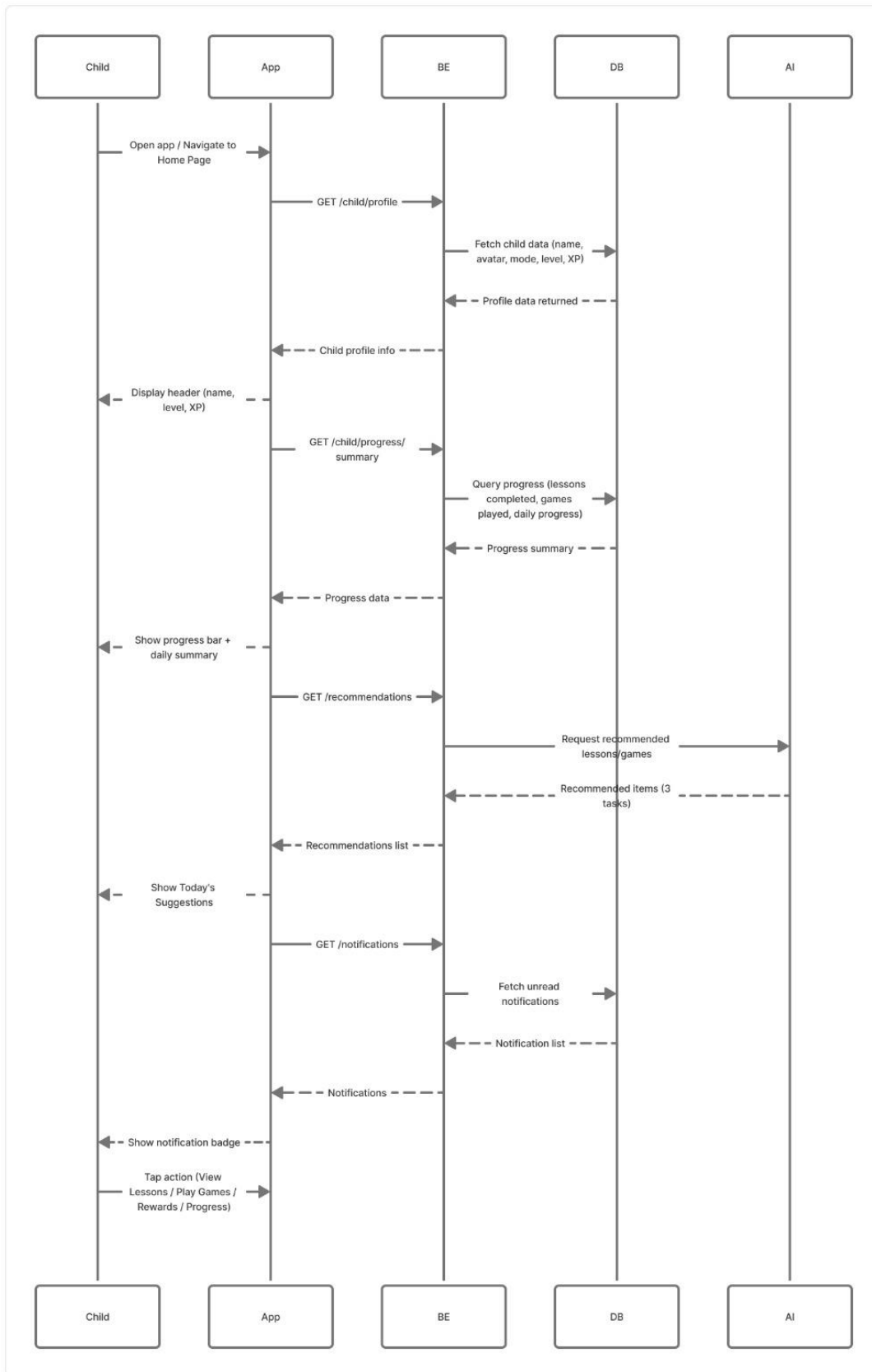
Sequence Diagram

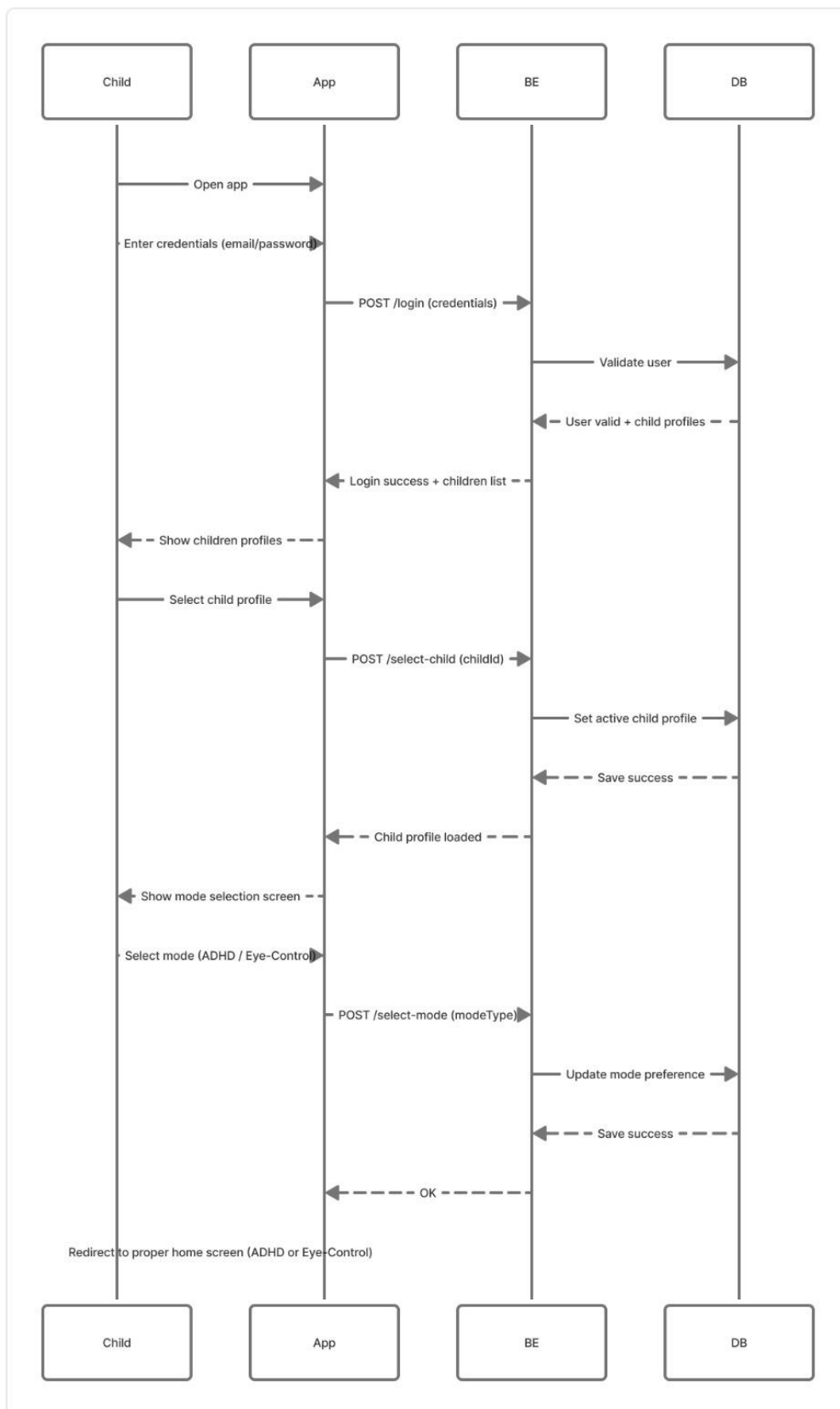


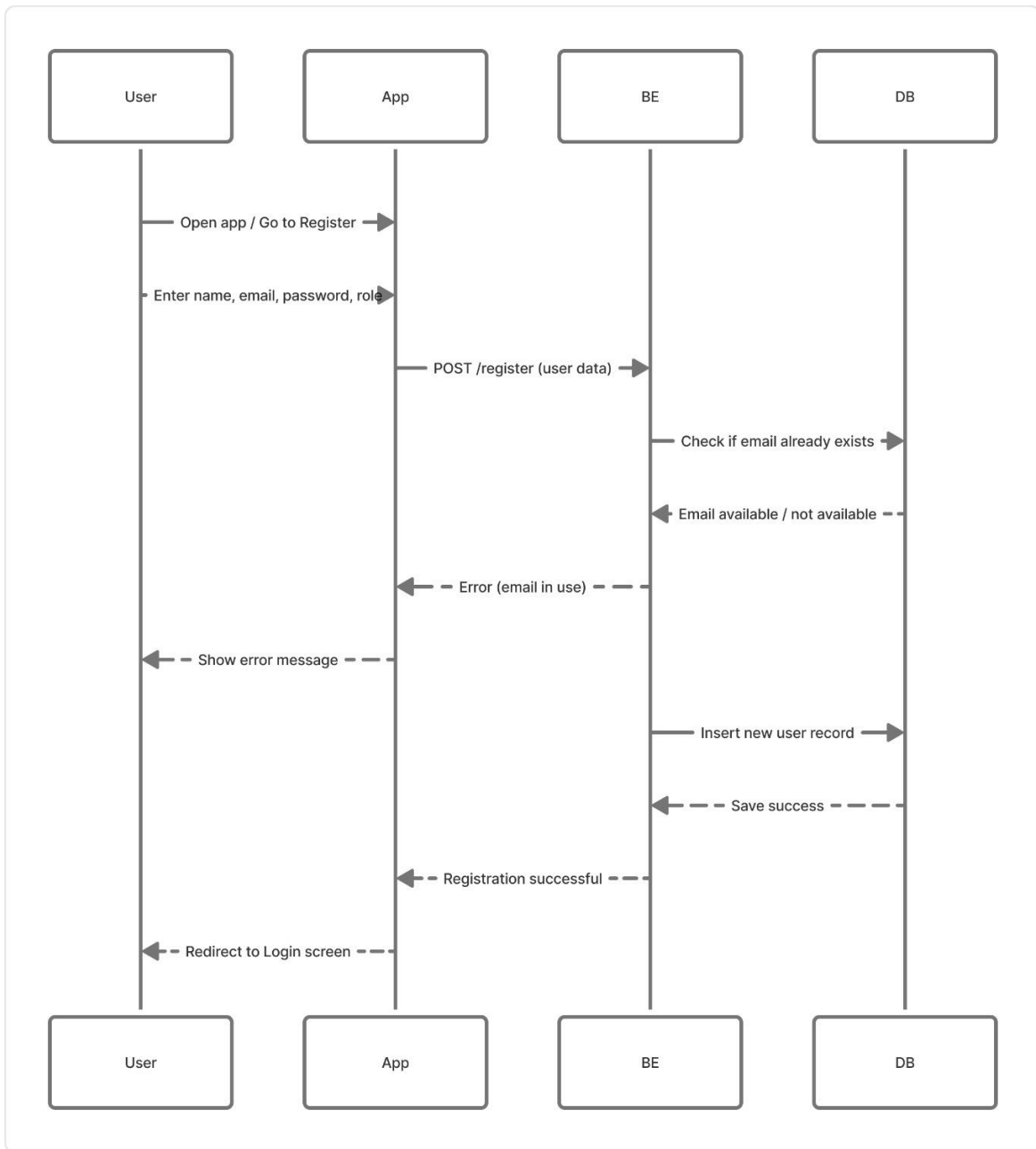


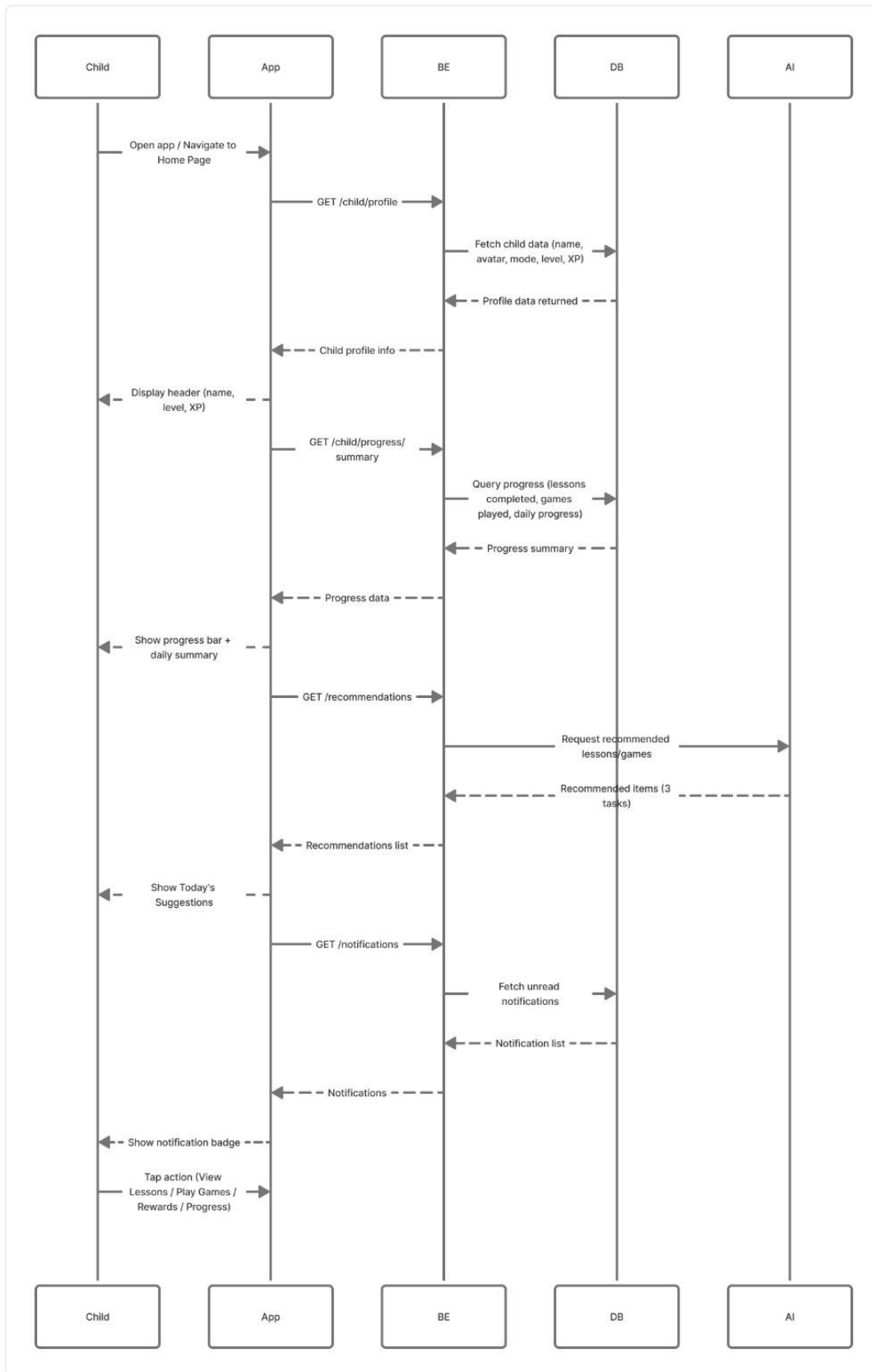




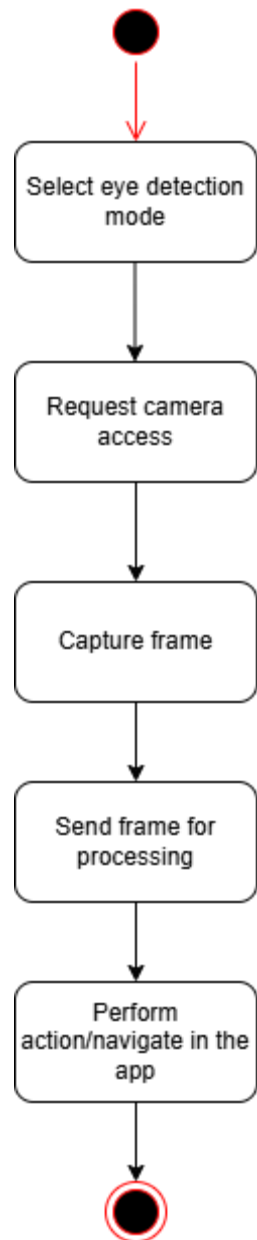


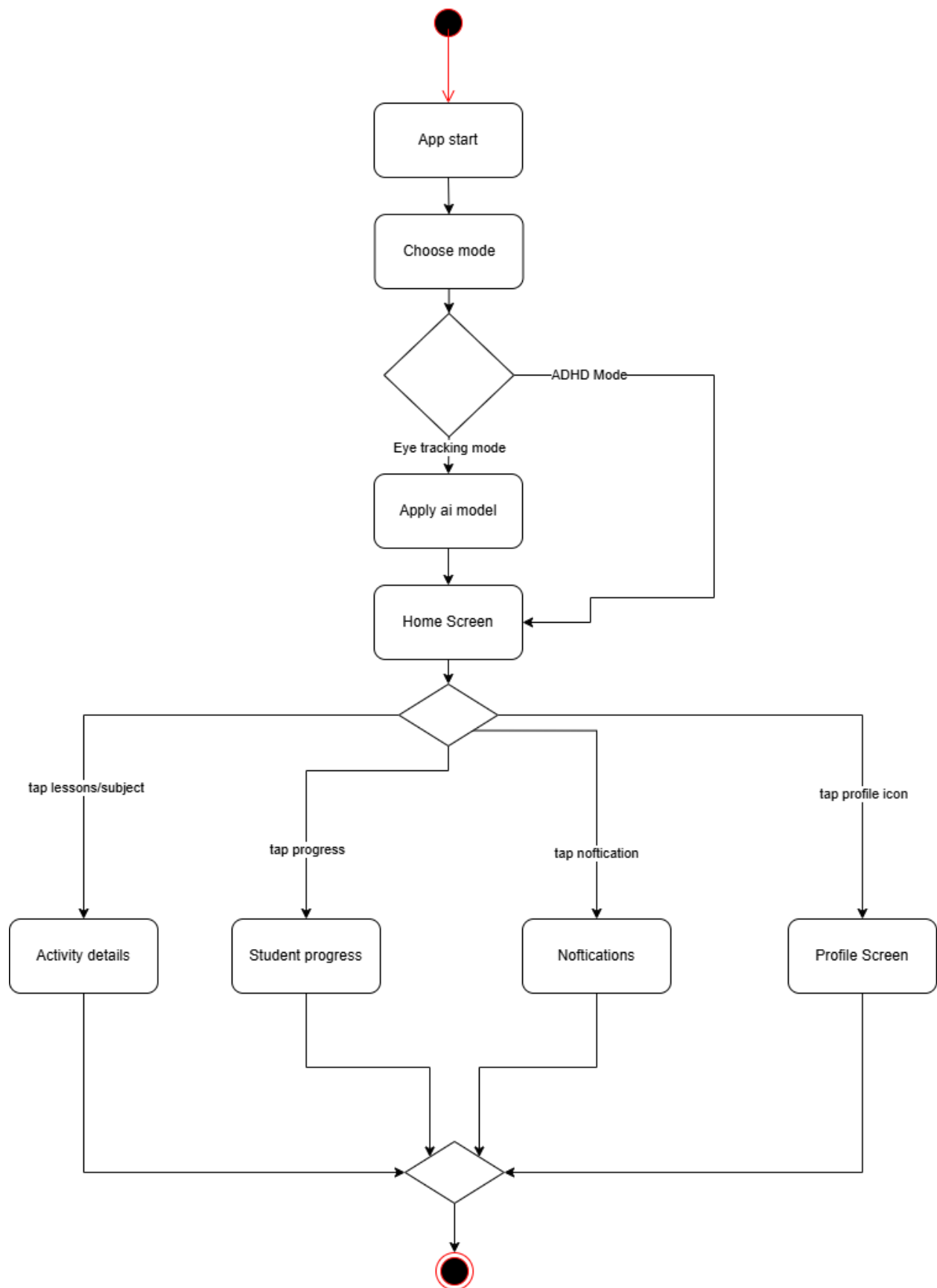


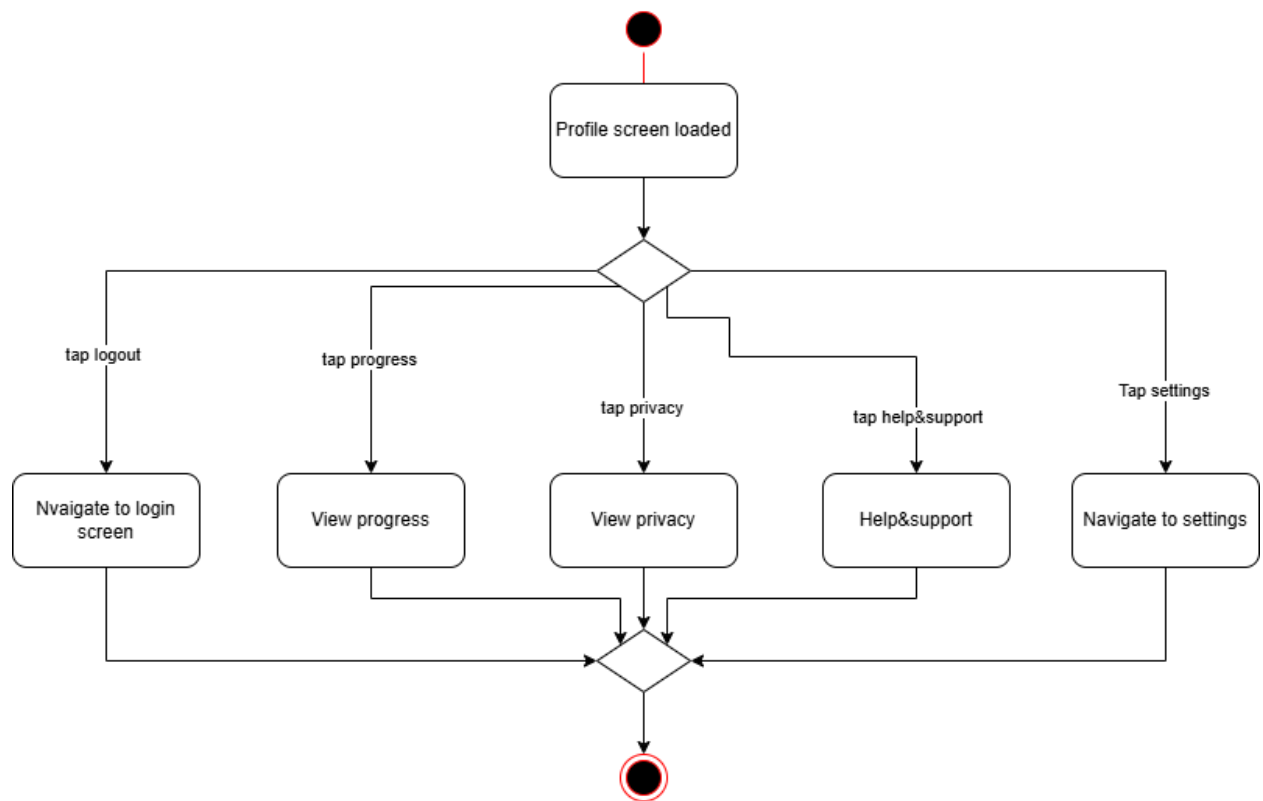


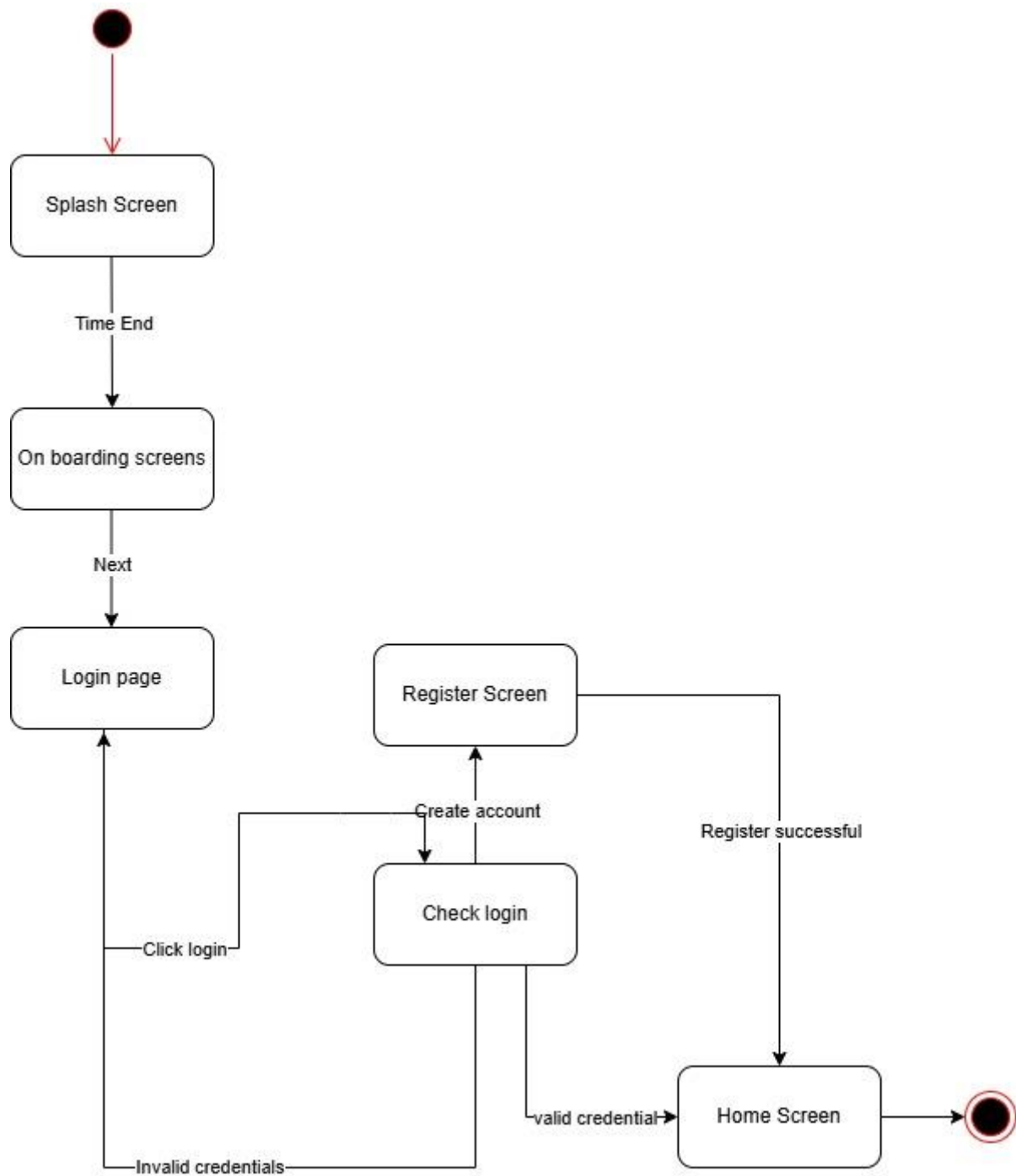


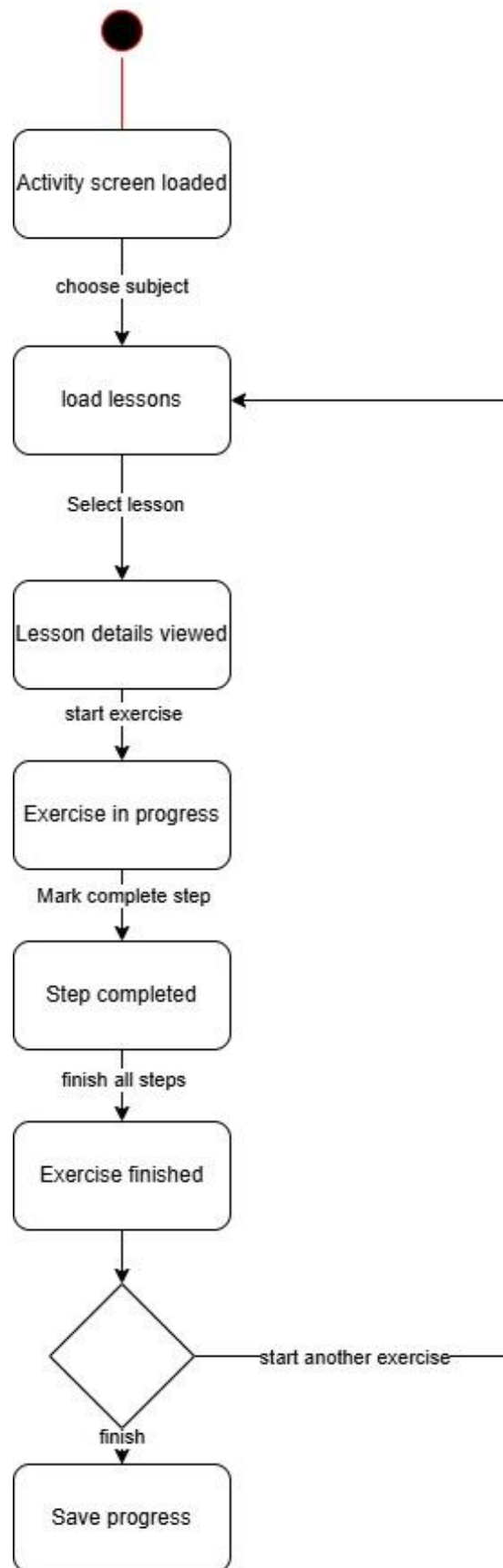
State Diagram



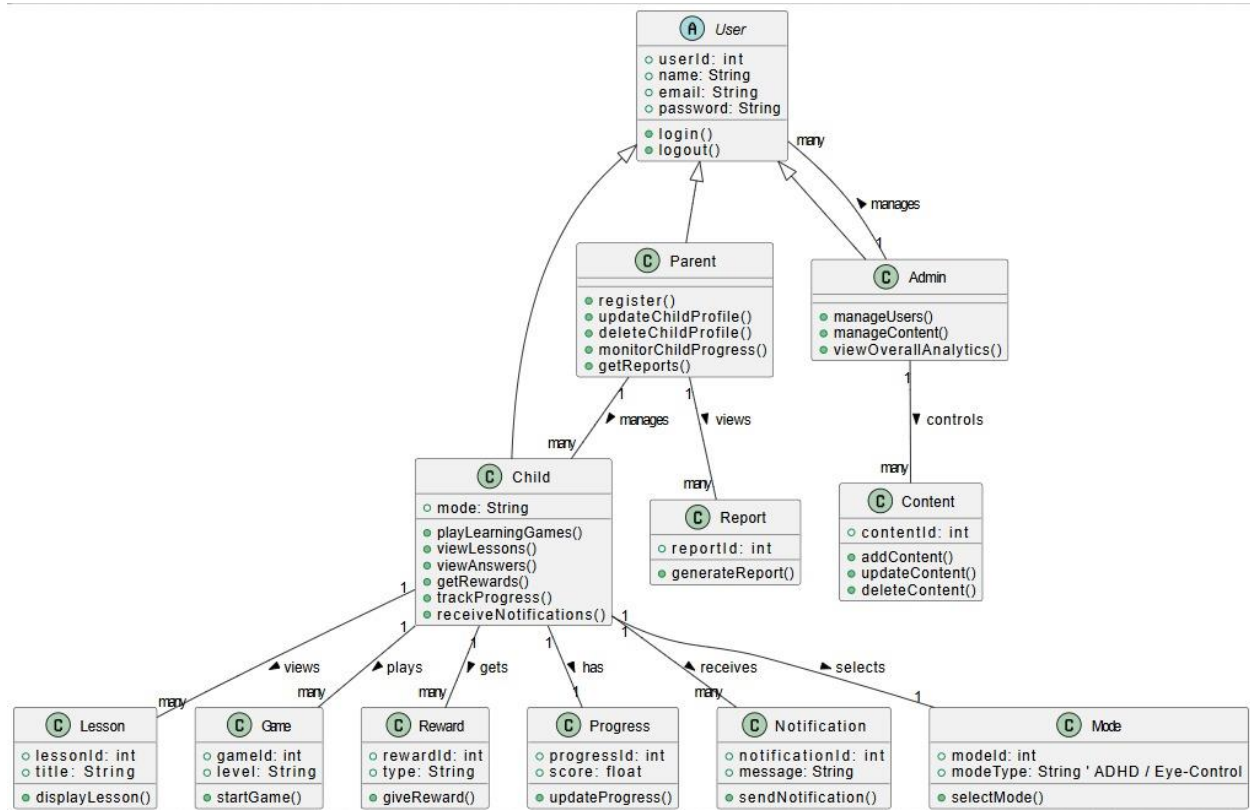




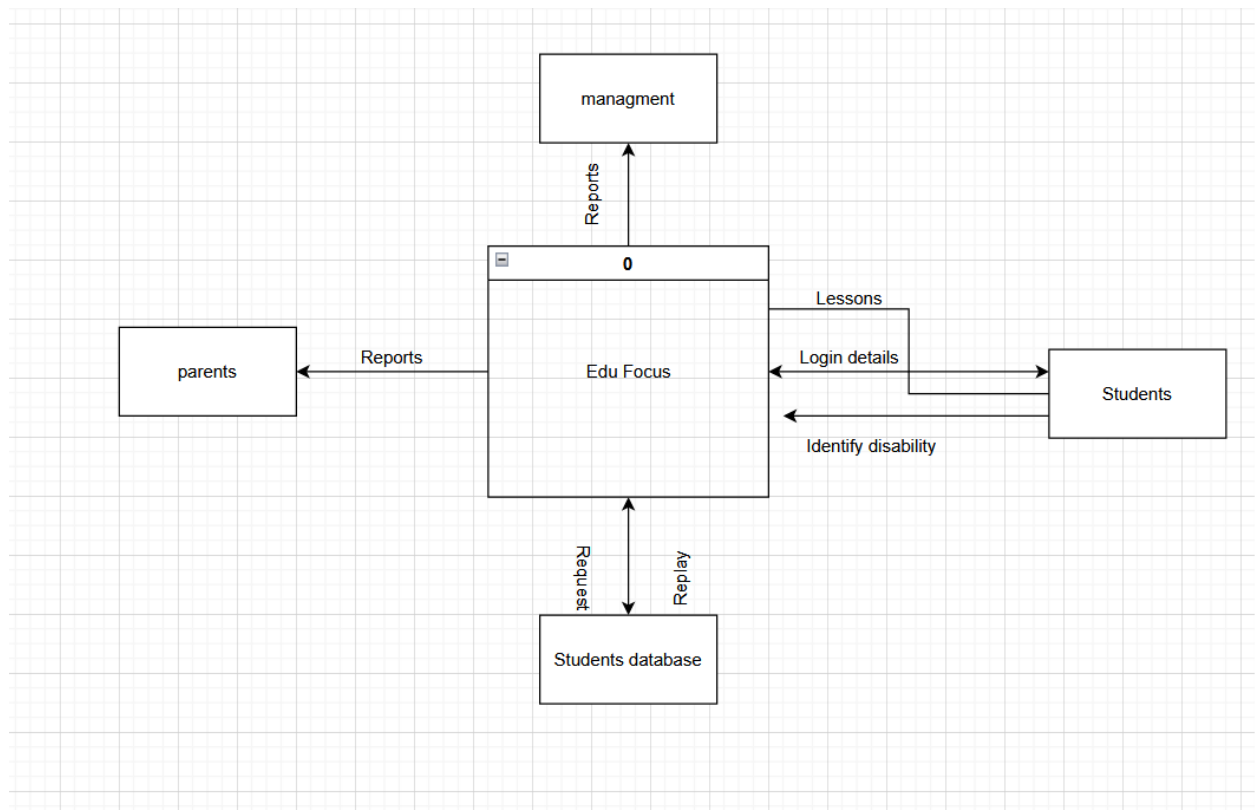




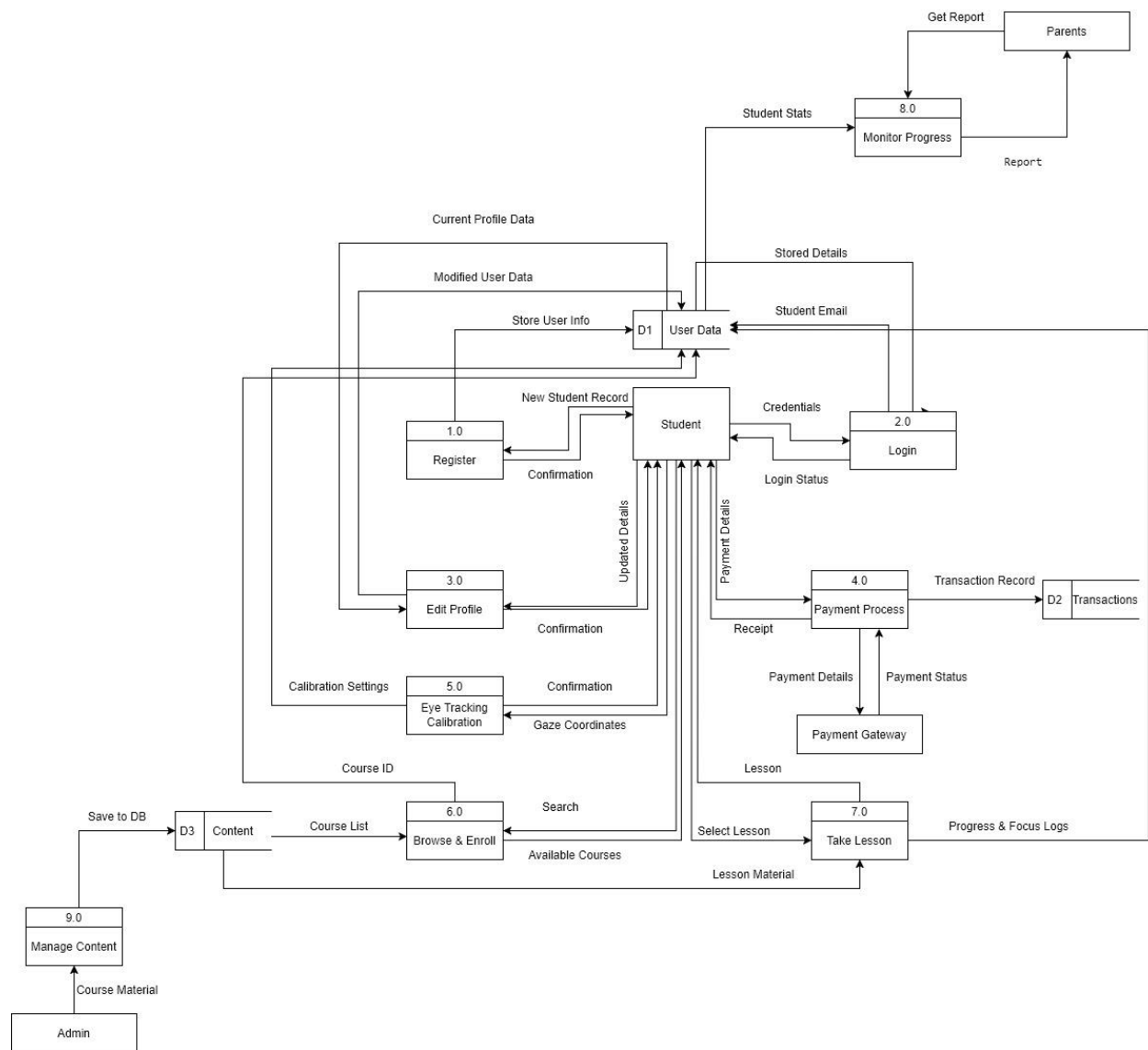
Class Diagram



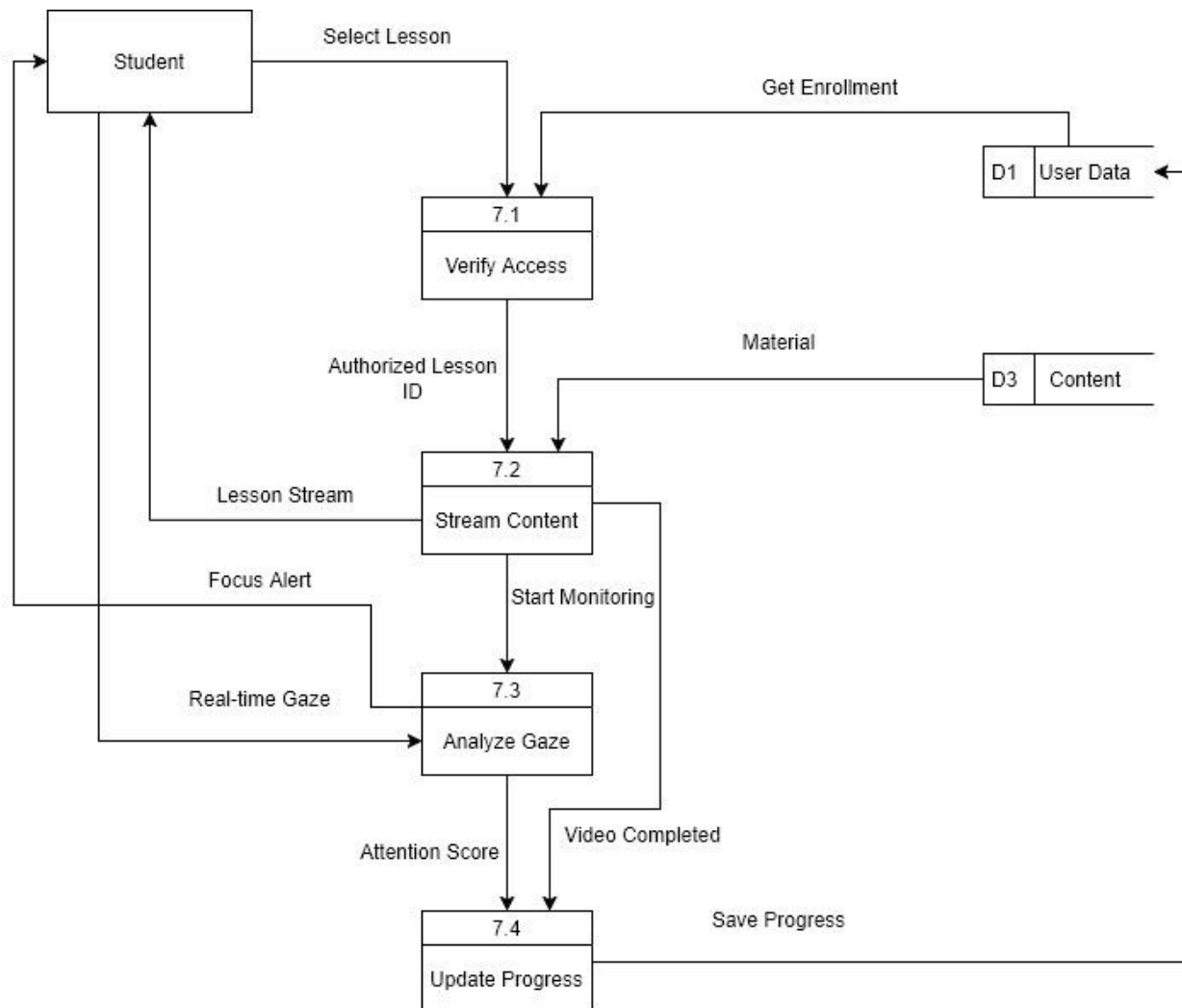
Context Diagram



DFD lvl-1

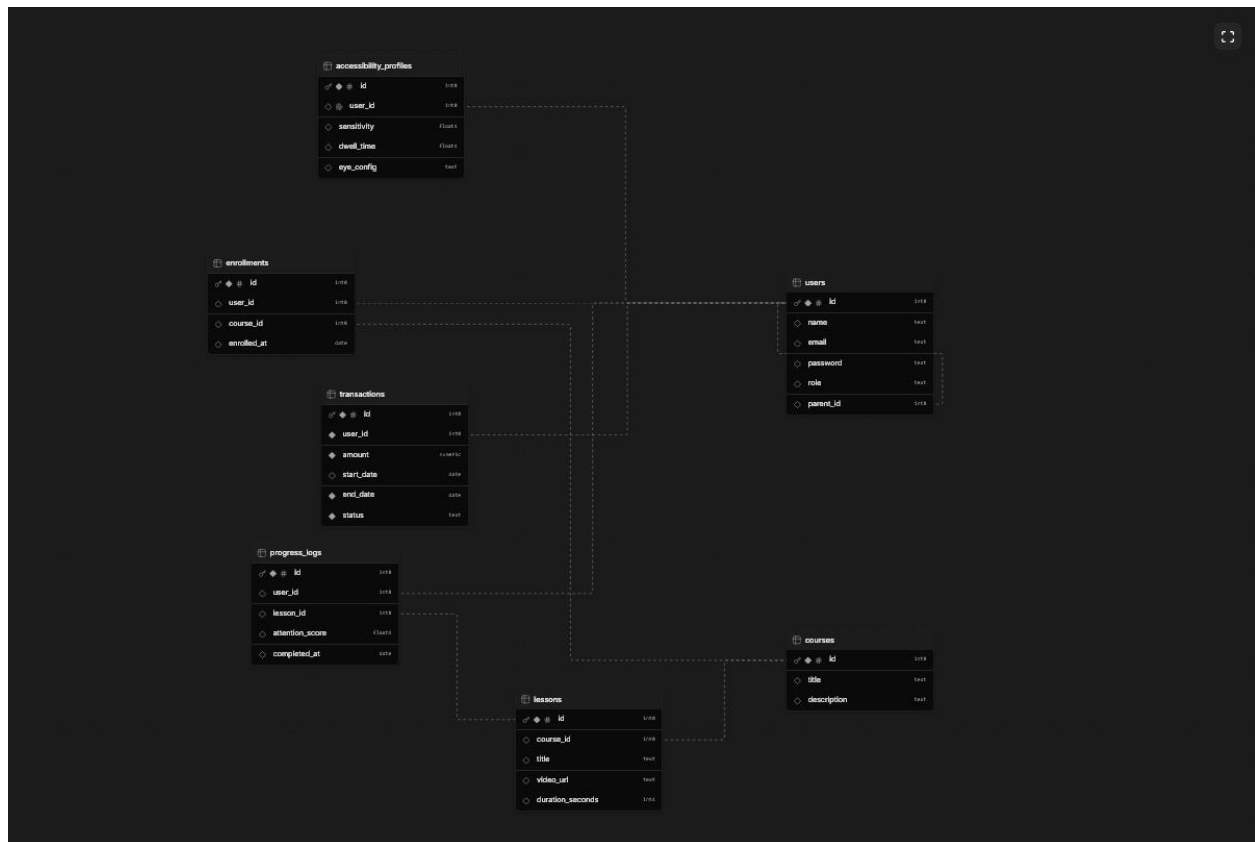


DFD lvl-2



Database Design

Schema Diagram



ERD Diagram

